RumiCar

RumiCar Hands-On!

Take on the Challenge of Self-Driving Algorithms

Lecturer: Rumika CHIBA
RumiCar Development Team
Project Owner

Text Version 2.4

RumiCar Meet the Team


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Shoko ISHIGAKI


Shigeki INADAMA

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Blue items involve programming

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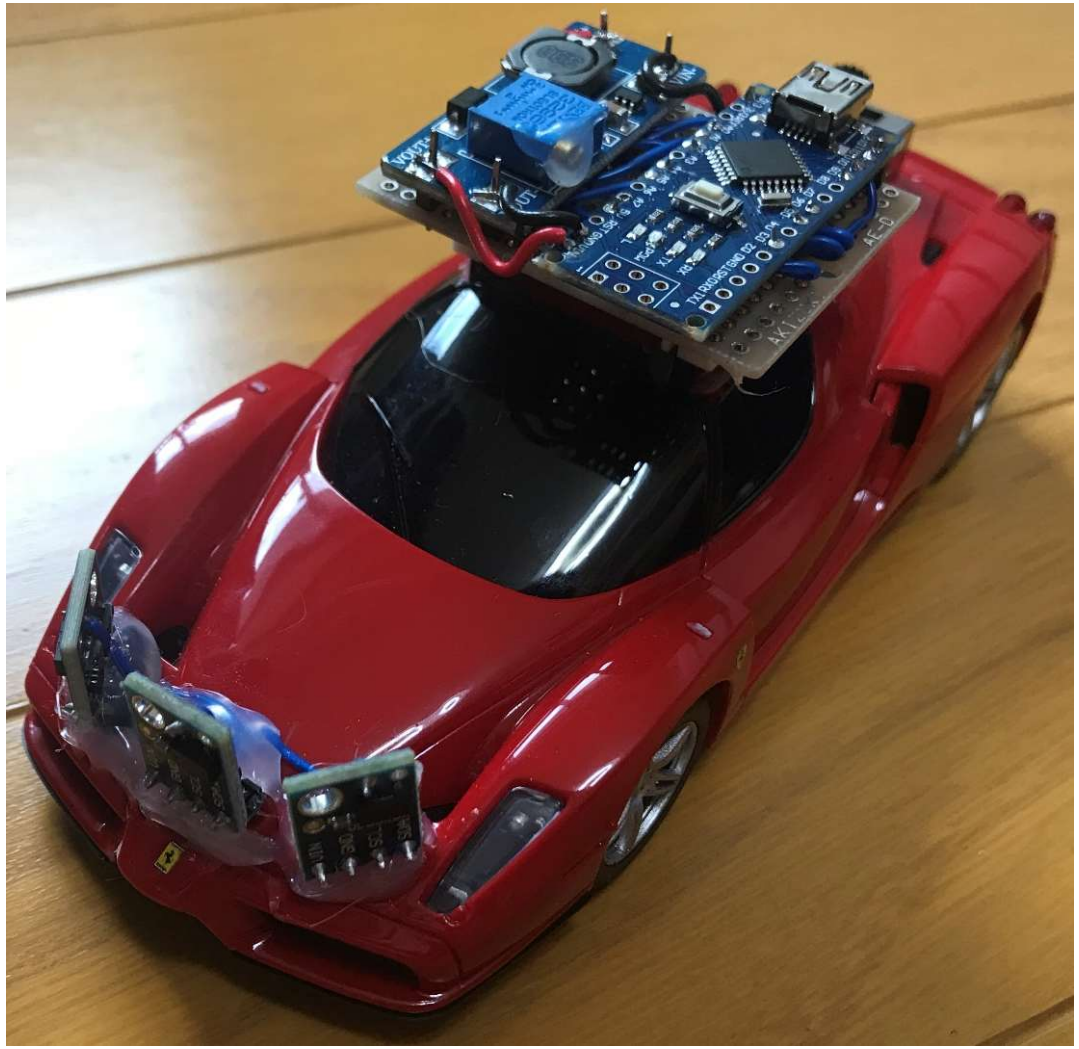
What You Will Learn

- How RumiCar Works
  - Sensors
  - Motor Driver
- Distance Measurement with Laser Ranging Modules
- Motor Driver Control
- Autonomous Driving Basics

- You will also learn why turning code into functions matters for good program development

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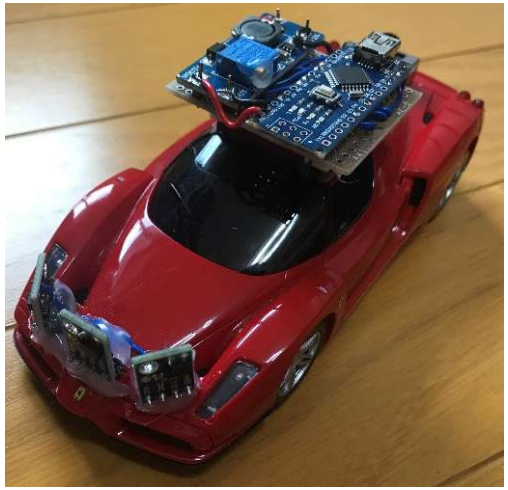
Introducing the RumiCar



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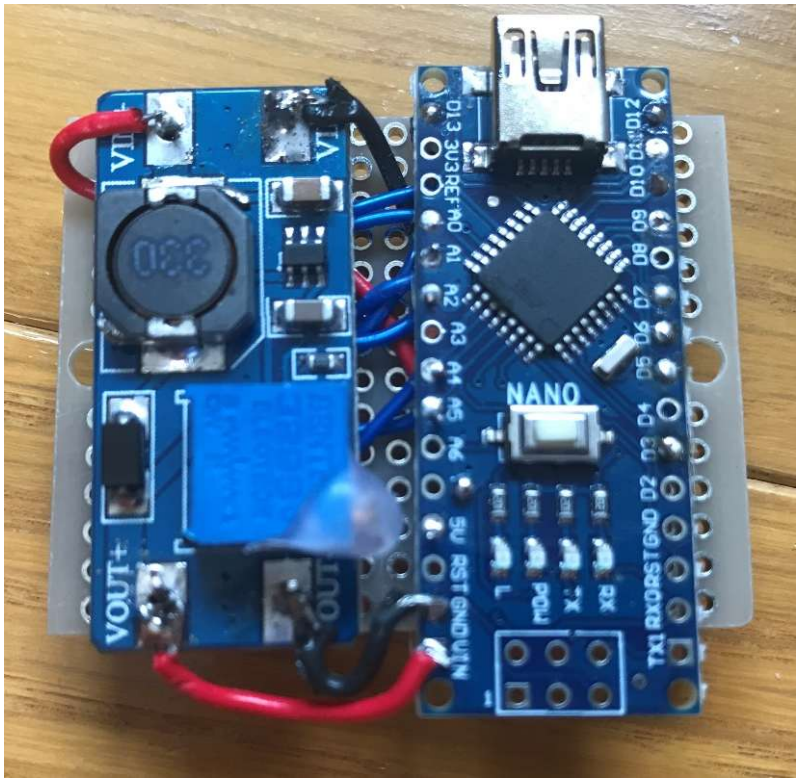
Introducing the RumiCar

- A platform for developing autonomous driving programs by combining multiple car bodies and various types of compute modules
- Equipped with laser ranging modules that can measure the distance to obstacles
- Vehicle speed can be controlled by controlling motor rotation

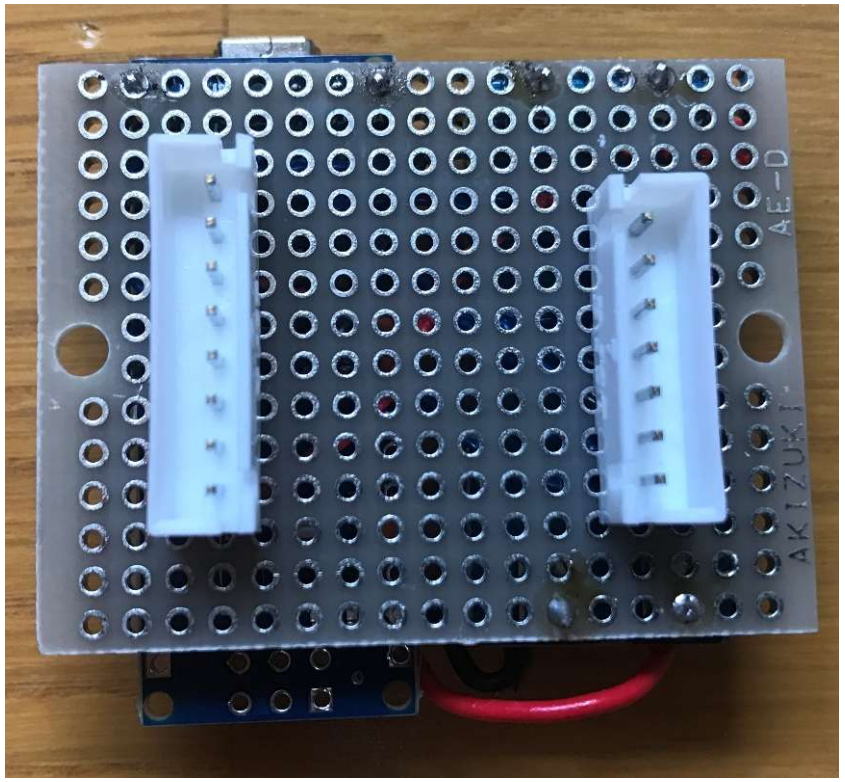


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Compute Module (CM)



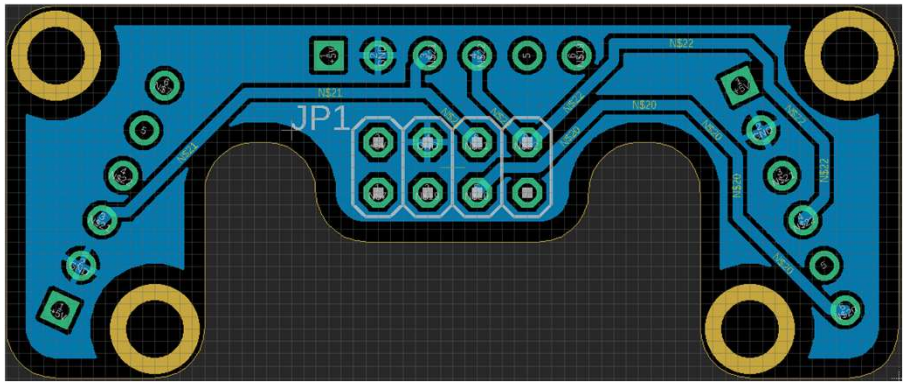
Front



Back

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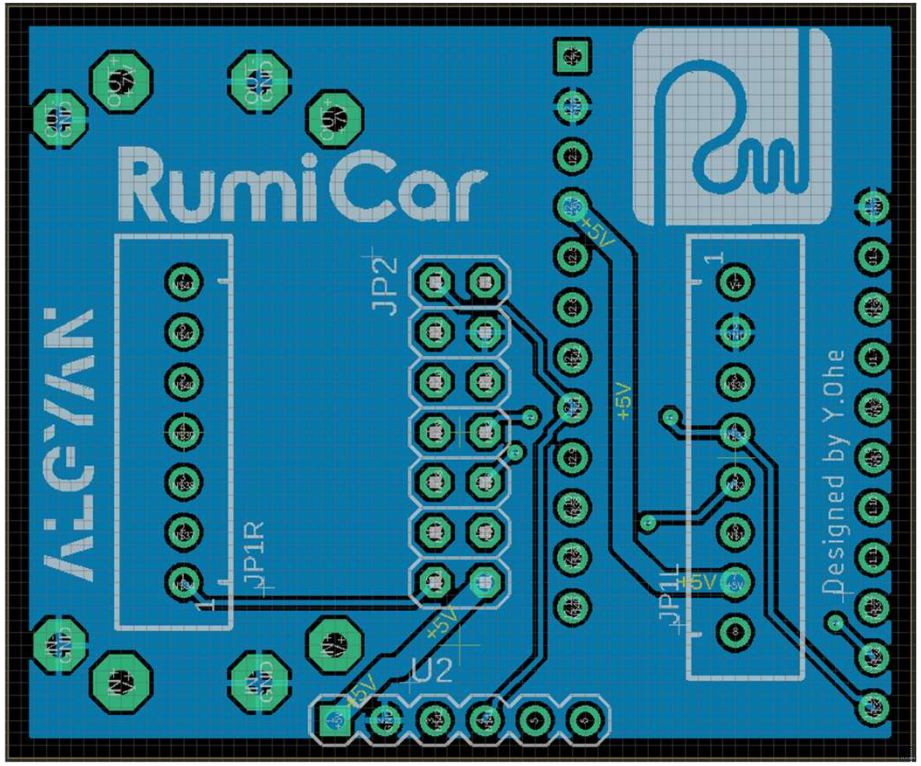
Compute Module (CM) on a PCB



Sensor PCB

Developed and designed by Mr. Yutaka OHE





Ordered from Seeed Fusion PCB.
Delivered.

Compute Module PCB

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Mix and Match Freely

Arduino Nano



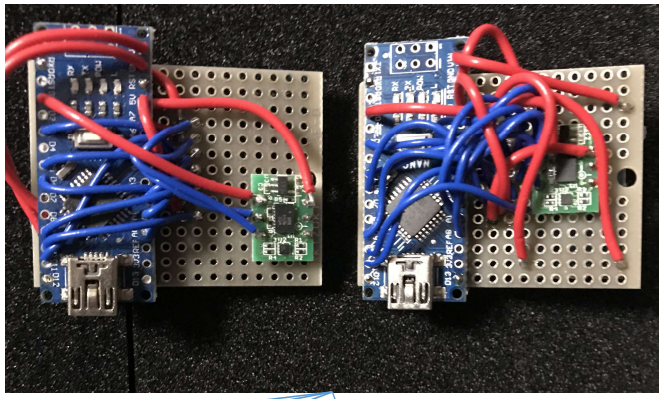
ESP32

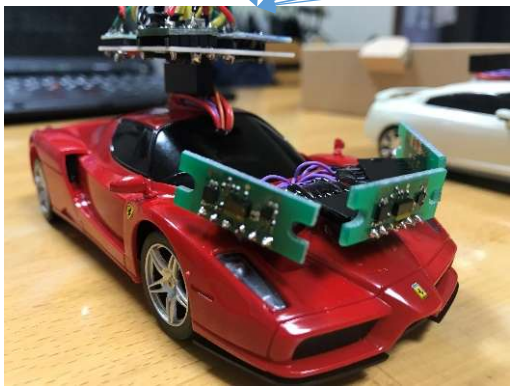


ESP32+EasyVR3



Arduino Nano+BMX055









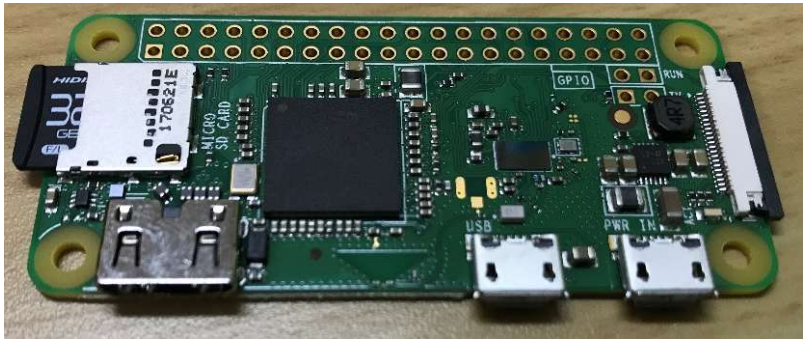


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Mix and Match Freely

Next Initiatives (Challenges)
Raspberry Pi

- **Compute module**
- **Motor control**
- **Image recognition**





Writes code in Python and works on motor control and more: Ms. Shoko ISHIGAKI

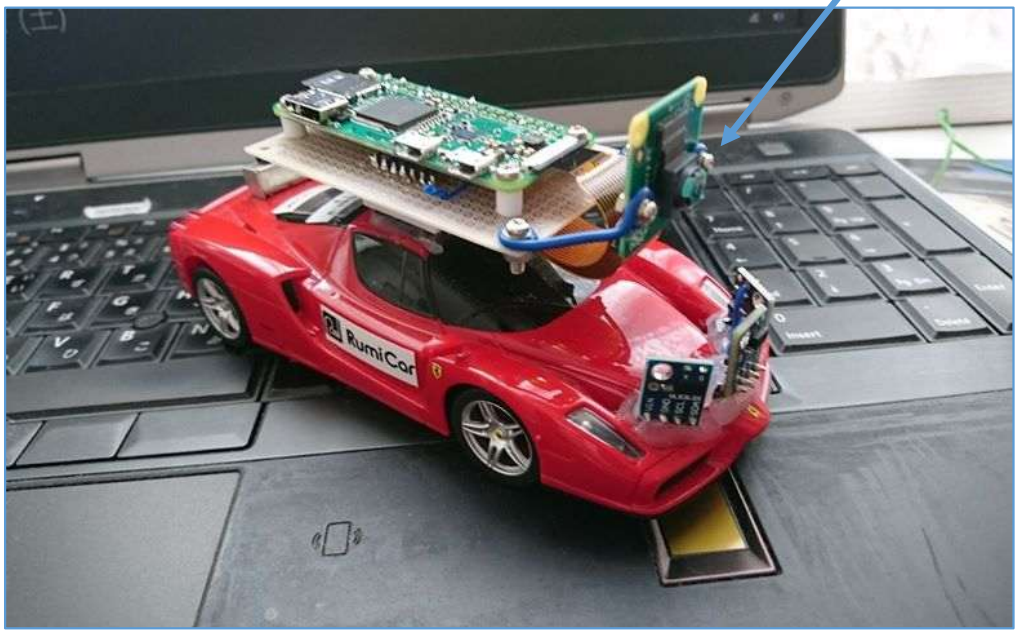
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Raspberry Pi CM + Camera



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Raspberry Pi CM + Camera



RasPi CM on a RumiCar



Taken with the RasPi CM camera

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Obniz CM



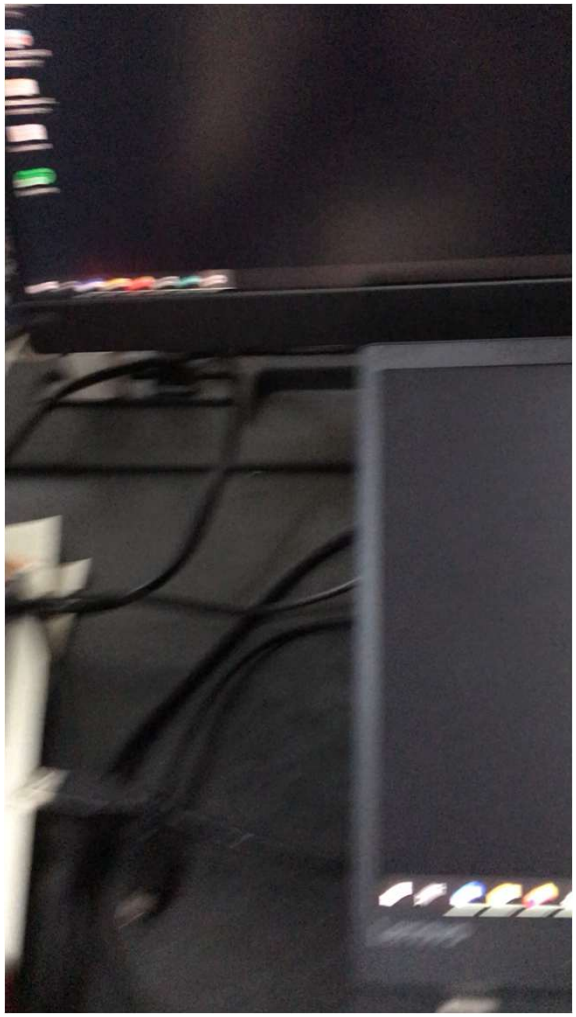
Left: Obniz version; right: Raspberry Pi version

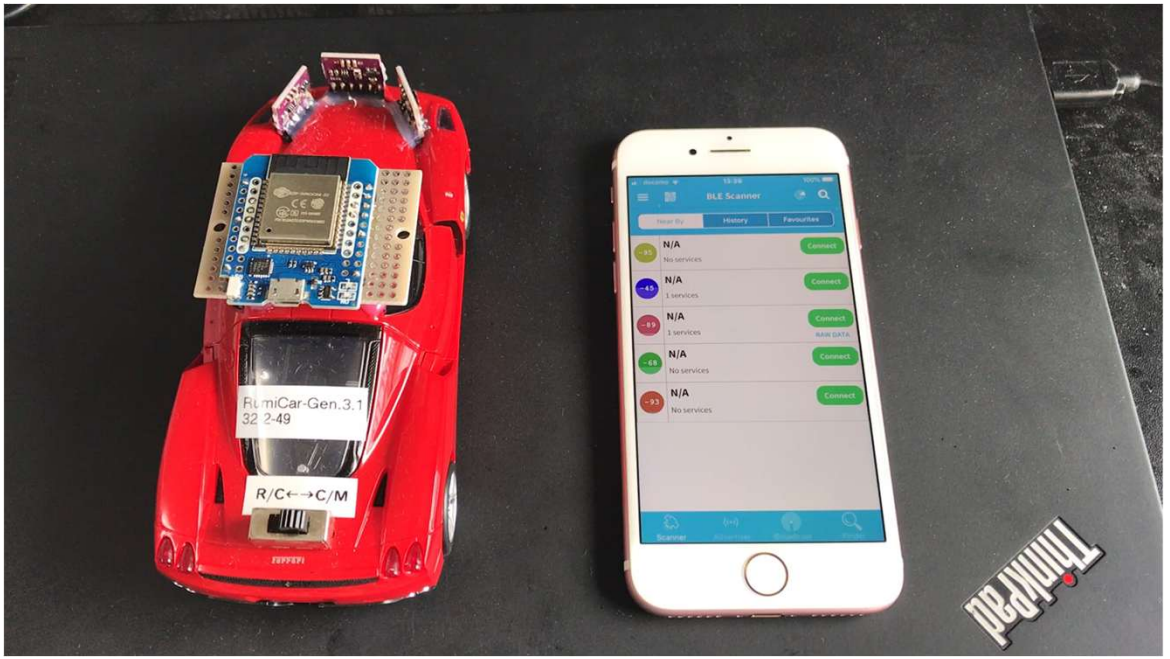
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ESP32 Bluetooth (Video)

https://youtu.be/-r6Vrhn5ifQ

https://youtu.be/UpEcTDvwYFw





BLE (Bluetooth Low Energy) multi-connect

Bluetooth serial connection

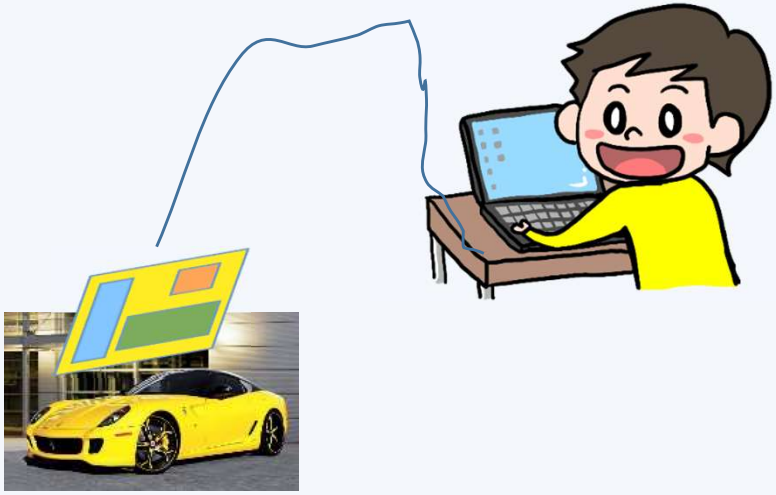
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Compute Module Comparison

Board	Arduino Nano	ESP32	Raspberry Pi Zero W	Obniz	Spresense
Power supply	7-12V(VIN) (Supplying the 5V line is not recommended)	2.2-3.6V (when input to 3V3) 3.7-6.0V (when input to VIN)	5V	5V	3.6-4.4V
Internal operating voltage	5V	3.3V	3.3V	3.3V	
I/O pin voltage	5V	3.3V	3.3V	5V (voltage selectable)	Switchable 5V / 3.3V
I2C voltage					1.8V
External power supply capacity	40mA per pin 100mA total	12mA max.	16mA per pin 50mA total	1A in 5V mode	6mA
Power consumption (current)	65mA	160-260mA with WiFi	125mA 270mA with WiFi	170mA average (WiFi not used)	
Number of PWM pins	6	4	2 (hardware)	6	
Power configuration	Power: MT3608 VL/DRV: from the 5V line	PQ3RD23 (or LM2596)	Power: MT3608 (or TPS63020) VL/DRV:PQ3RD23 (or SOT-23 AP7333)	MT3608 (or TPS63020)	

Have Fun with Programming

Program it yourself!



- With a RumiCar you can program it from your own computer, and drive it freely
- Take on AI
- Write programs that will amaze everyone!

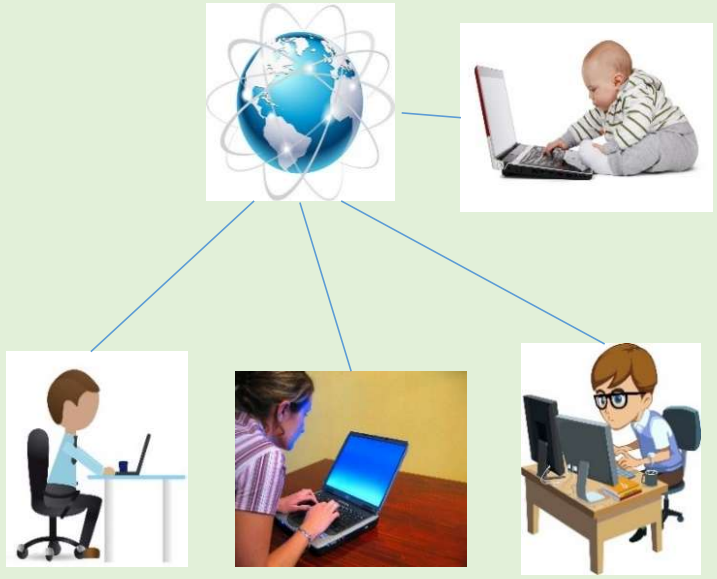
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Connect with Friends Worldwide!

Planned / in development

Download and exchange programs

- Upload your own programs to the server
- If you find an interesting program, download it to your own RumiCar and try driving it
- Connect with friends around the world

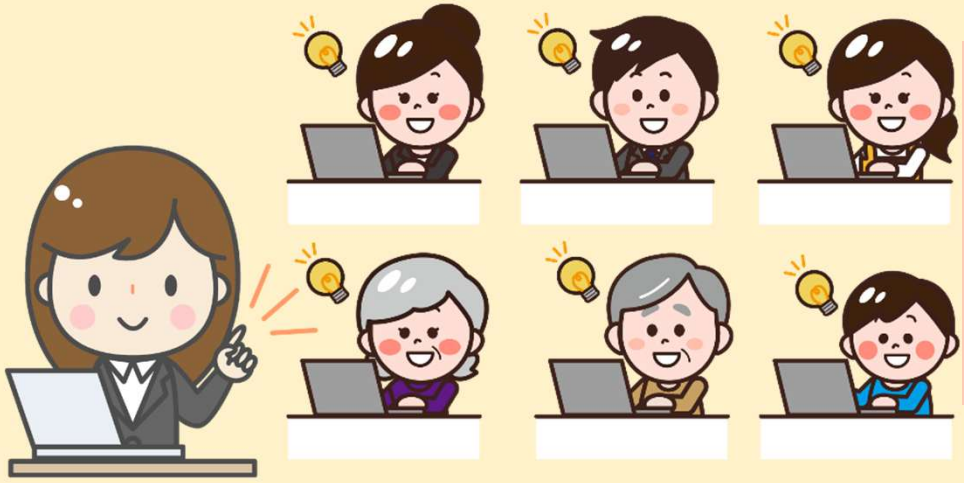


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Let's Learn Together!

Planned / in development

We hold classes



- Build a RumiCar!
- Build a compute module!
- Write programs!

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RumiCar Through the Years (Car No.1 to No.5)

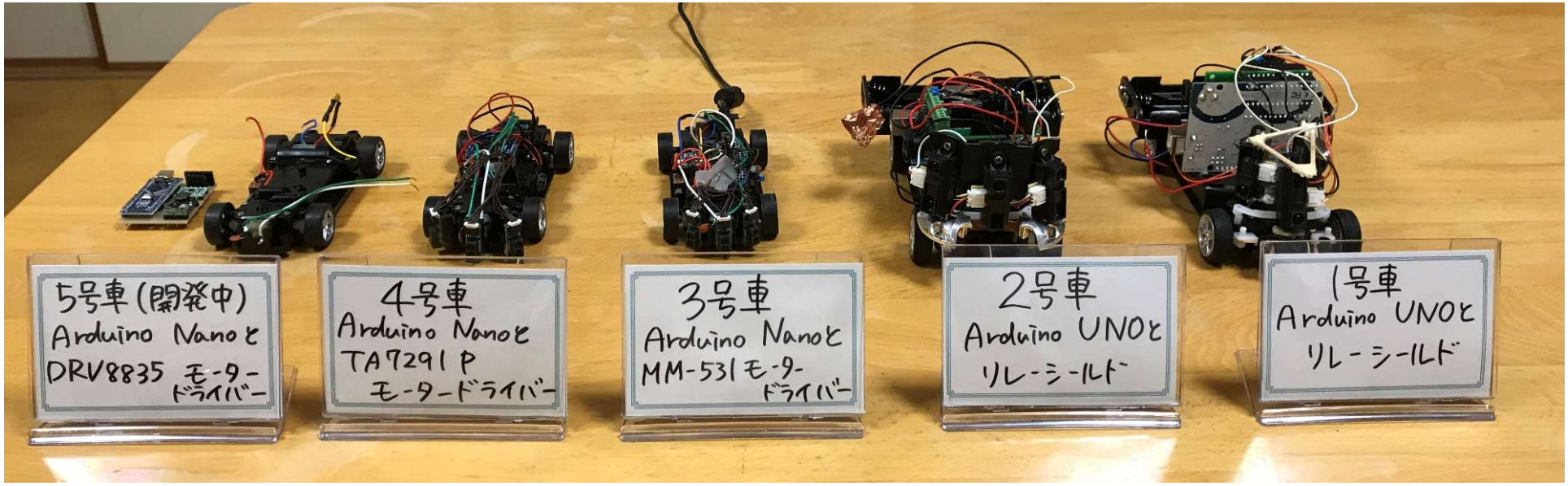
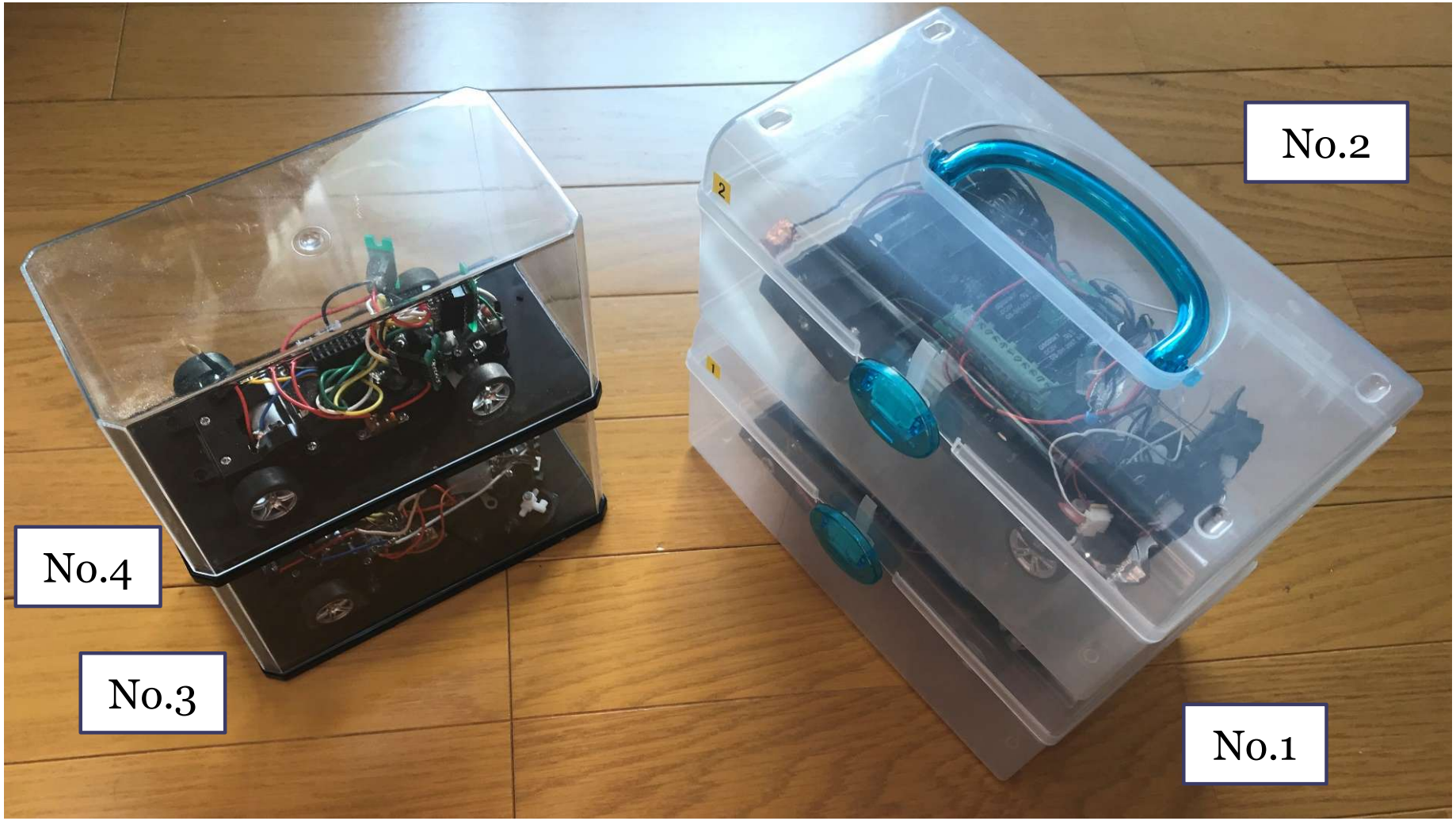


Photo taken on August 3, 2017

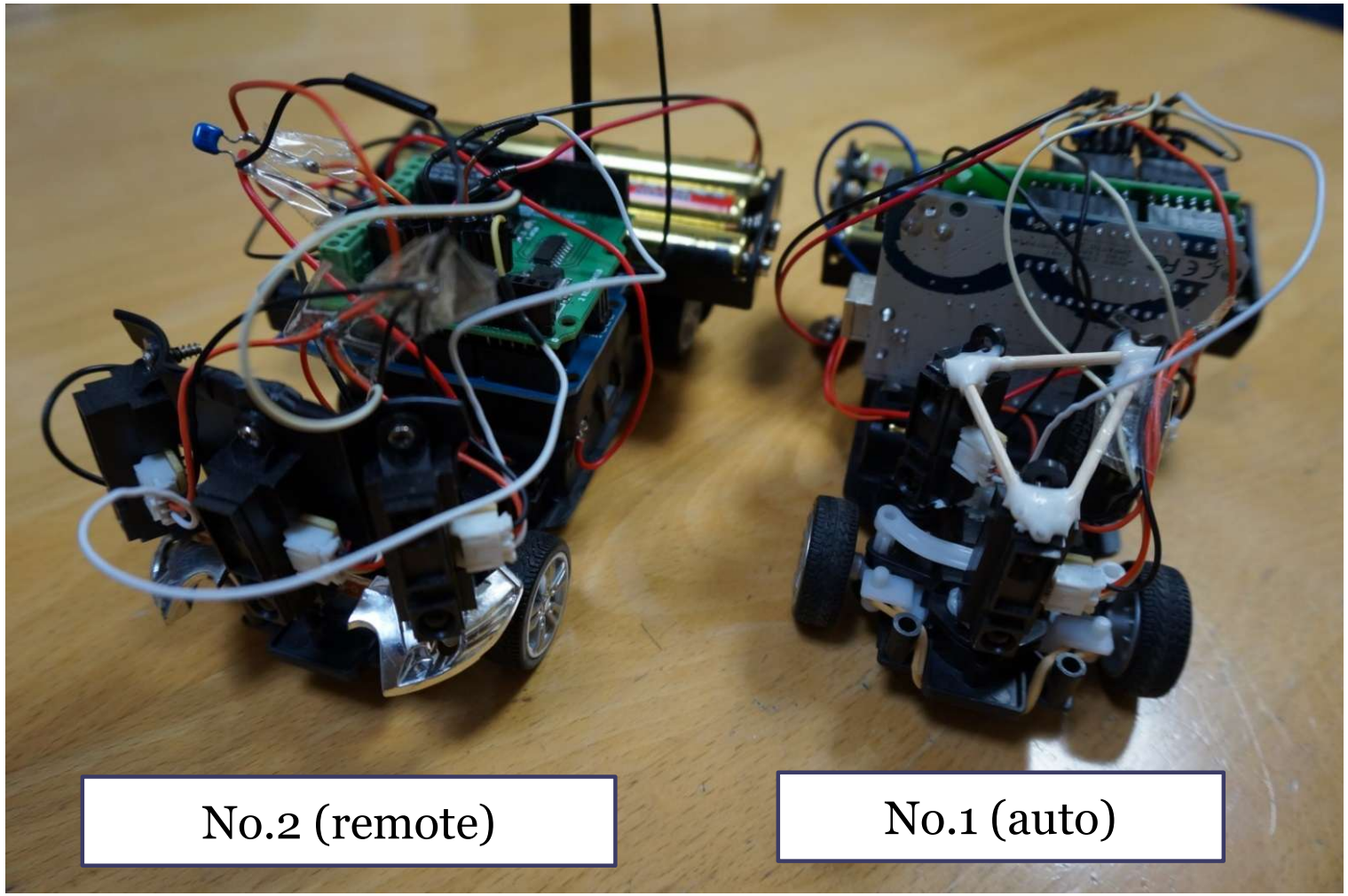
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RumiCar History (Museum)



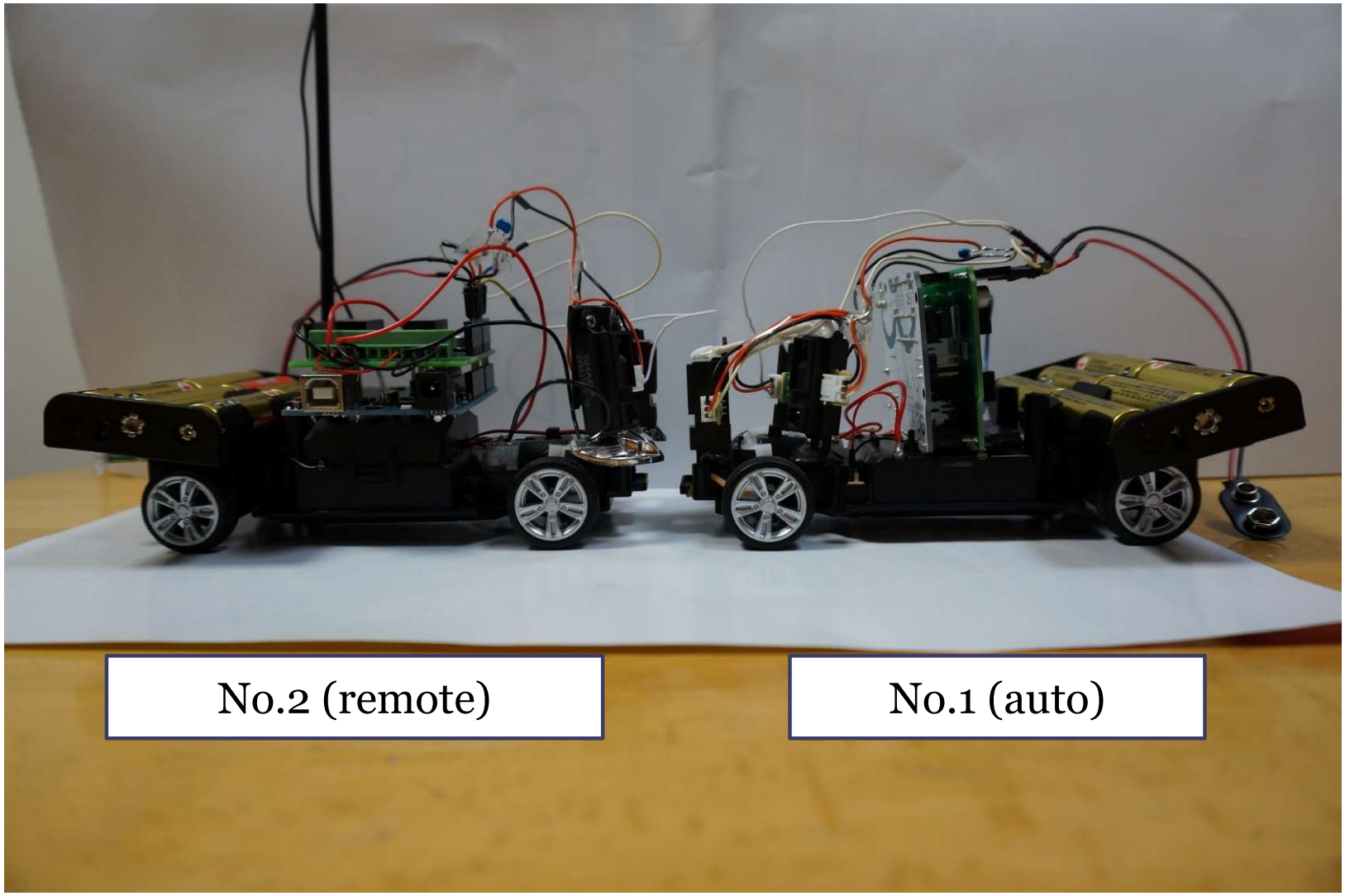
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RumiCar History (No.1 & No.2)



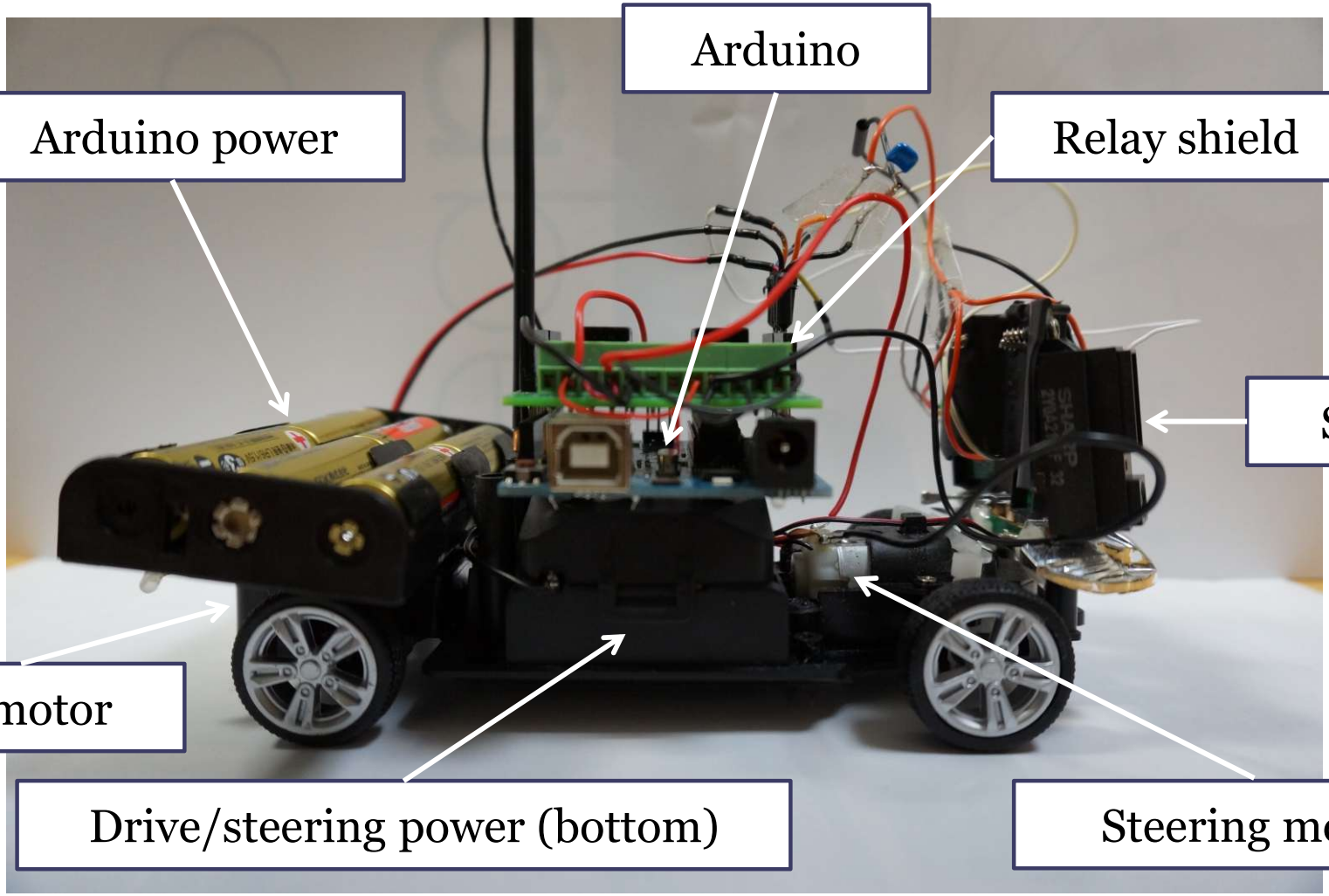
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RumiCar History (No.1 & No.2)



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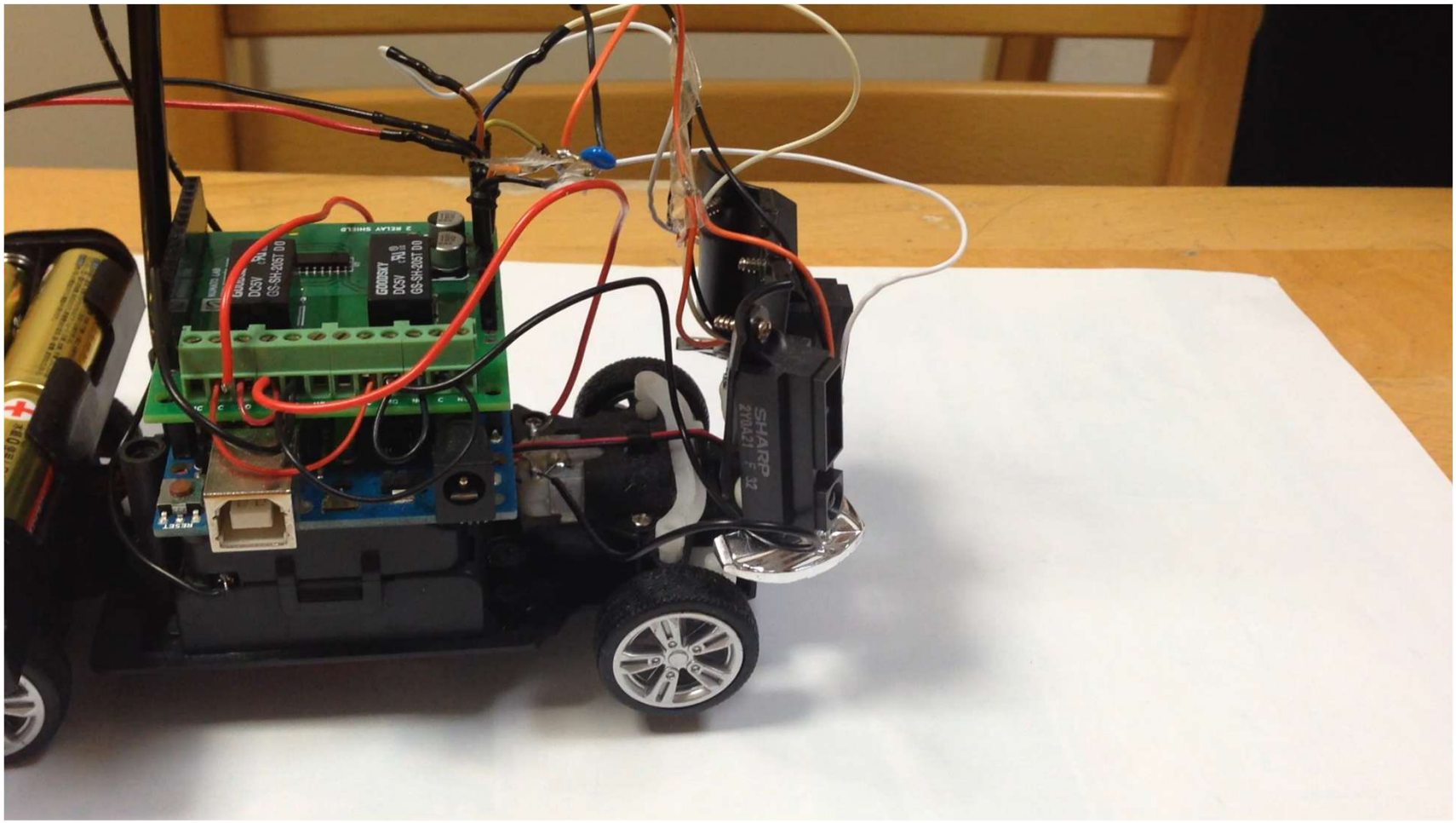
About Car No.2



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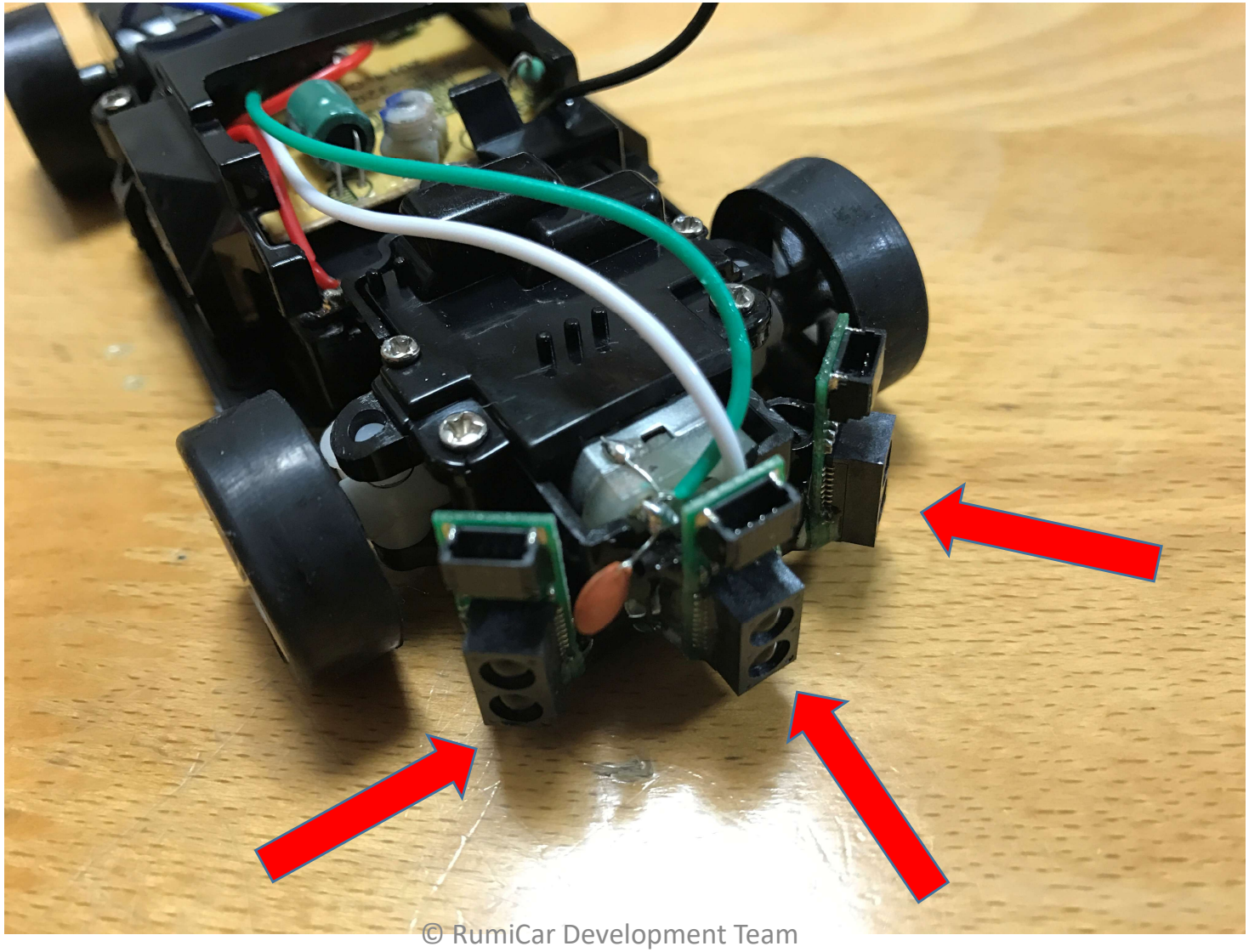
Sensor Test (Video)

https://youtu.be/vIAL7uUnXTA



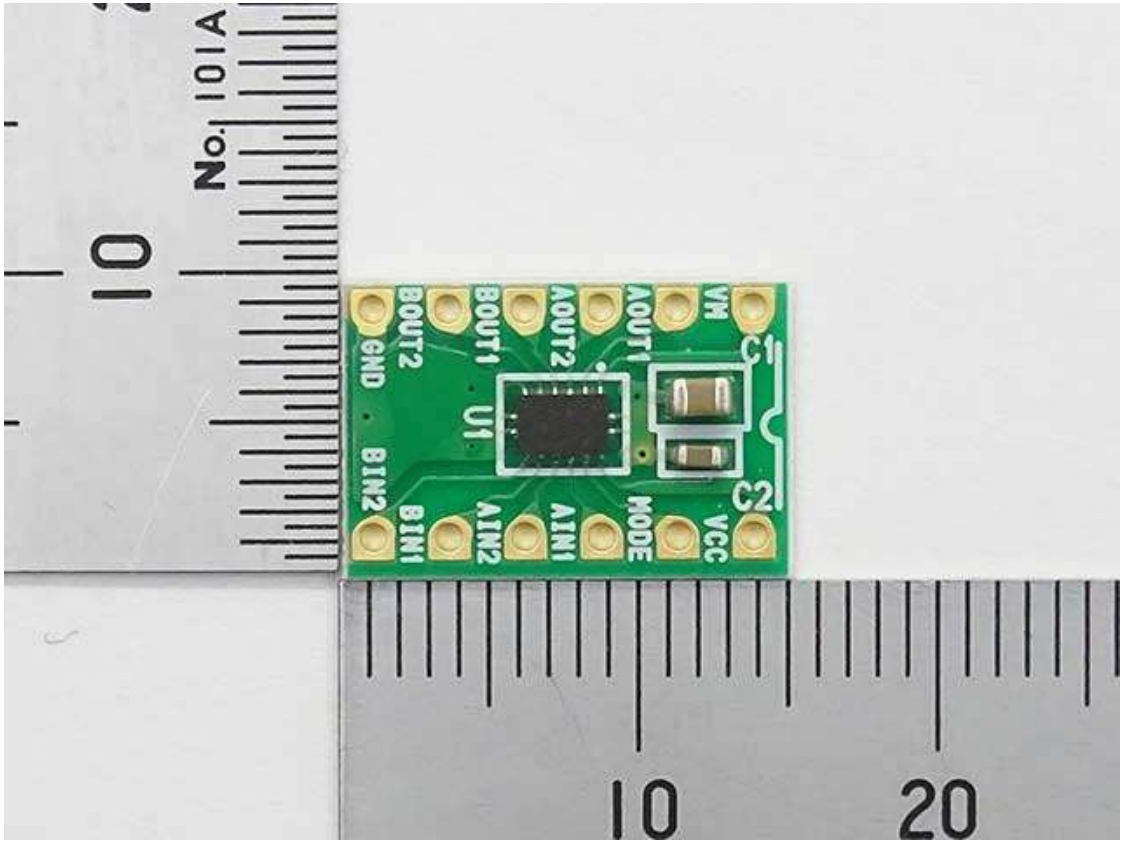
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Car No.3/No.4 Sensors



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Motor Driver



DRV8835

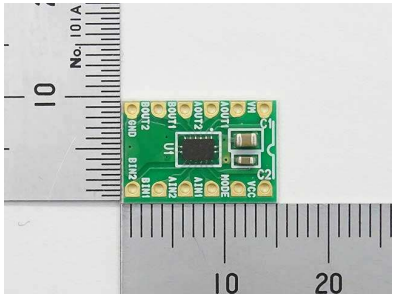
DRV8835 (from Akizuki Denshi Tsusho)

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Motor Driver

- The motor driver on the RumiCar is the DRV8835
- Features of the DRV8835
  - 2 channels
  - Up to 1.5A per channel
  - PWM(250kHz)
  - Low power consumption
  - Low voltage drop (motor side)
  - Logic power (VCC) and motor power are independent
    - Logic supply voltage: 2 to 7V
    - Motor supply voltage: up to 11V

DRV8835

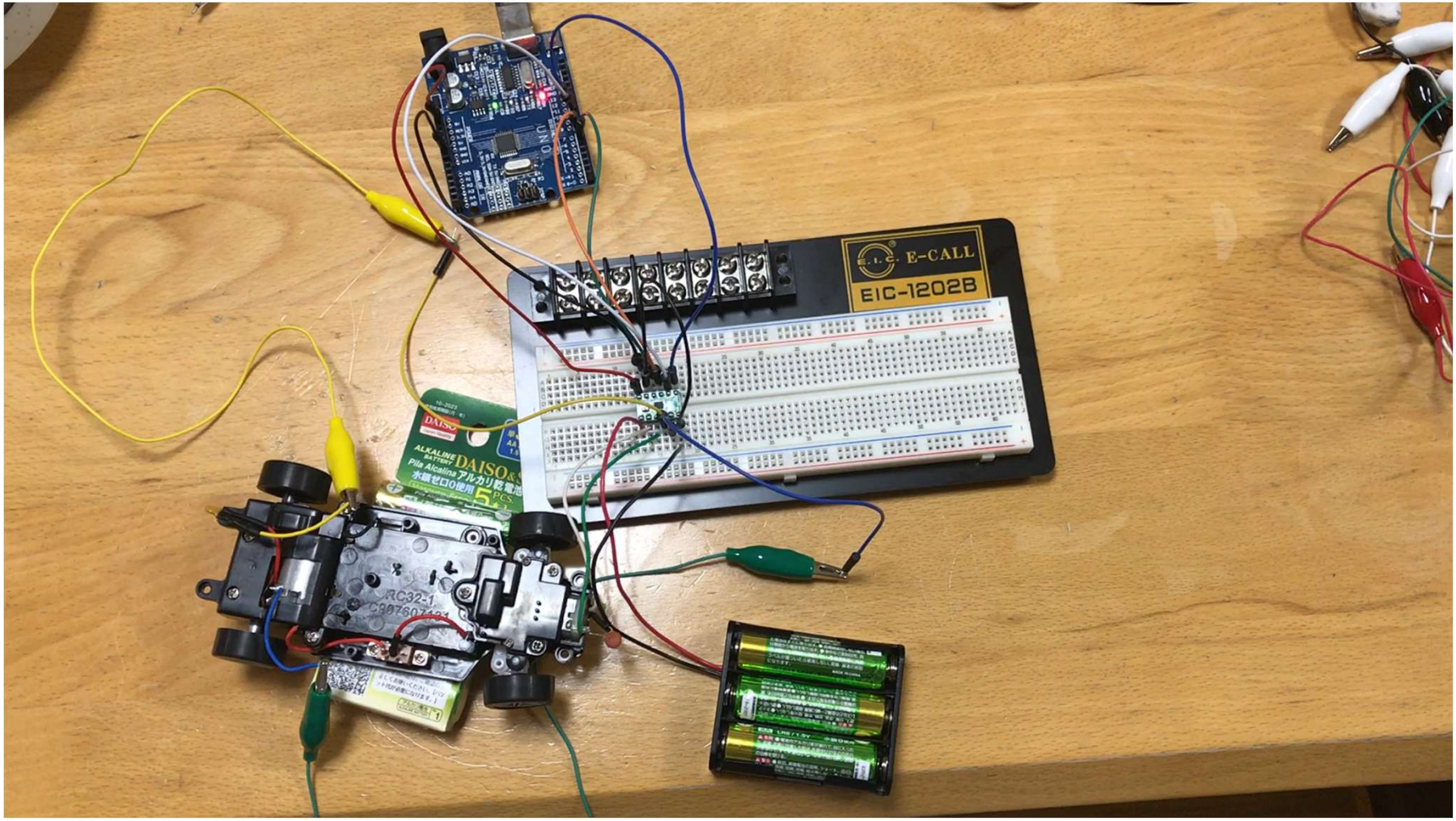


DRV8835 (from Akizuki Denshi Tsush

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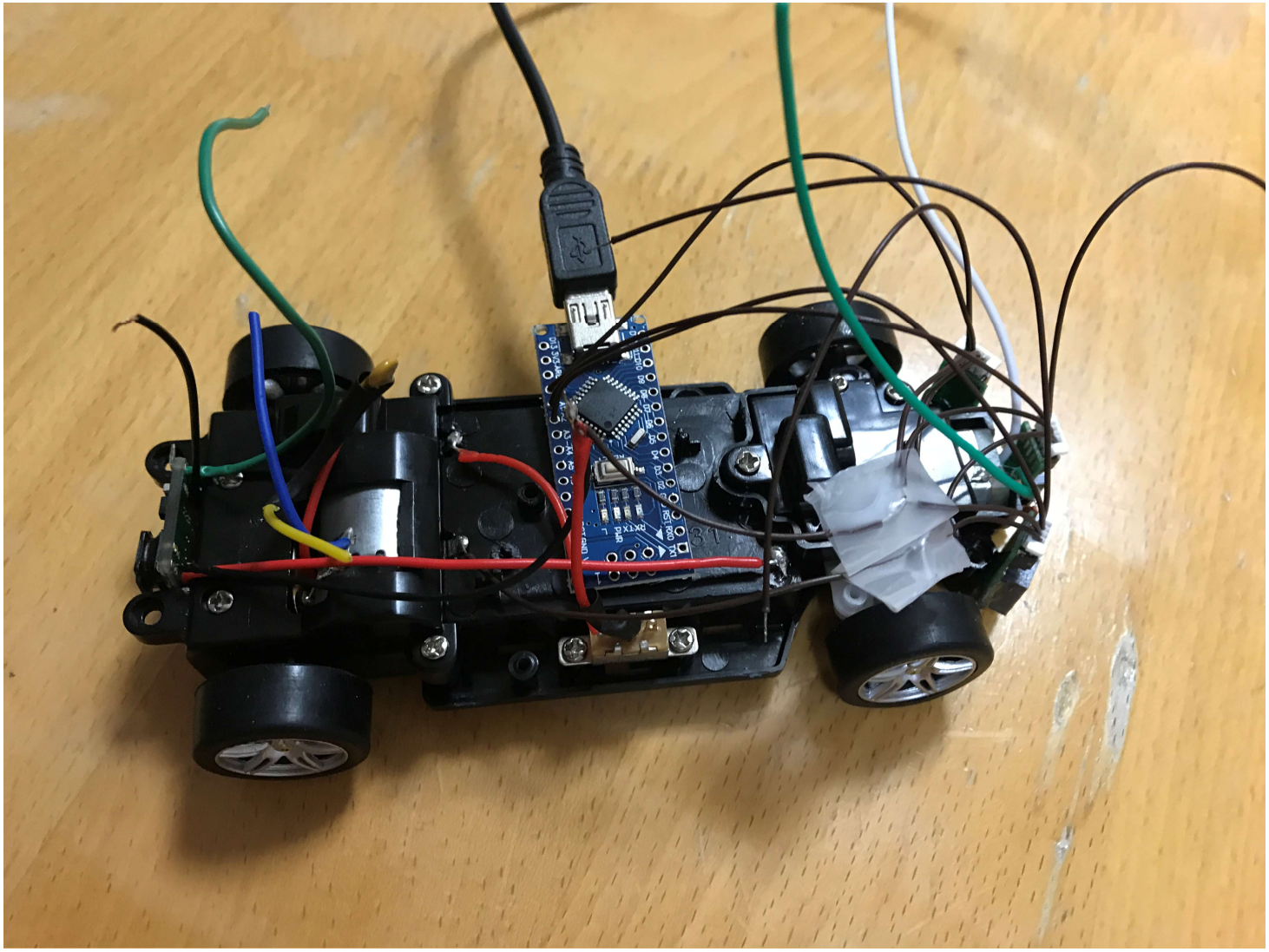
Development Video (Video)

https://youtu.be/8jULg3fZ-jw



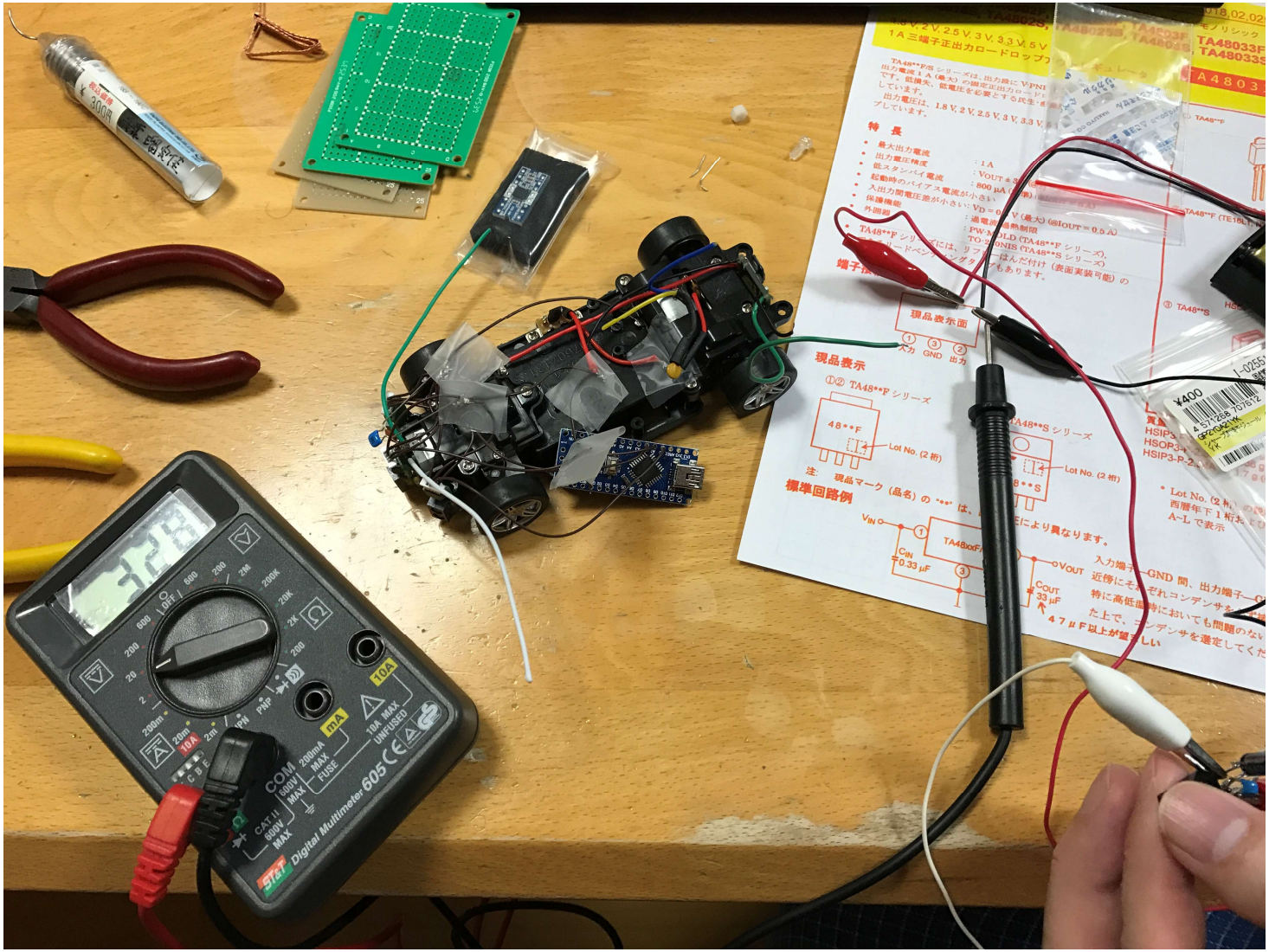
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Development Photo



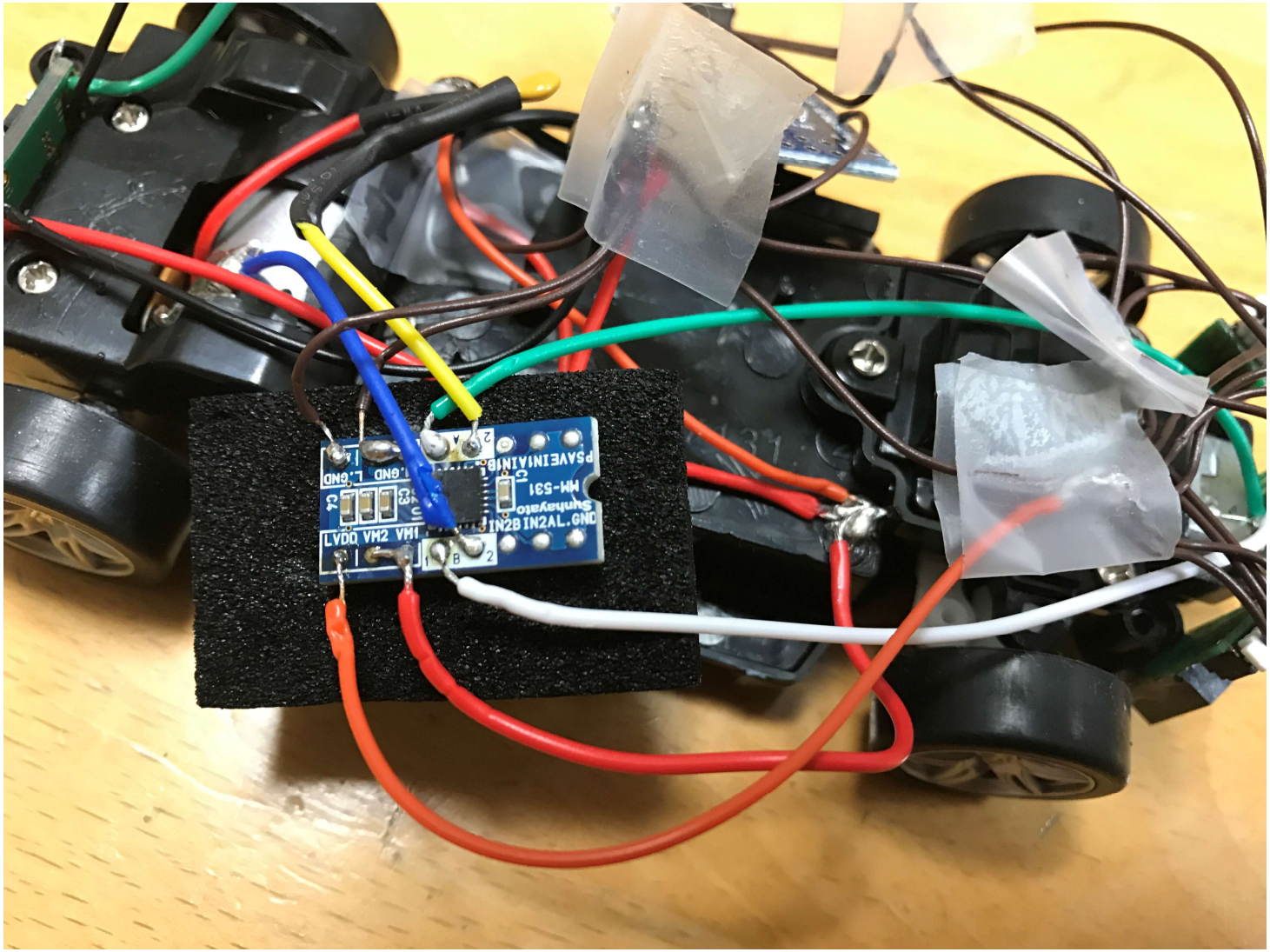
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Development Photo



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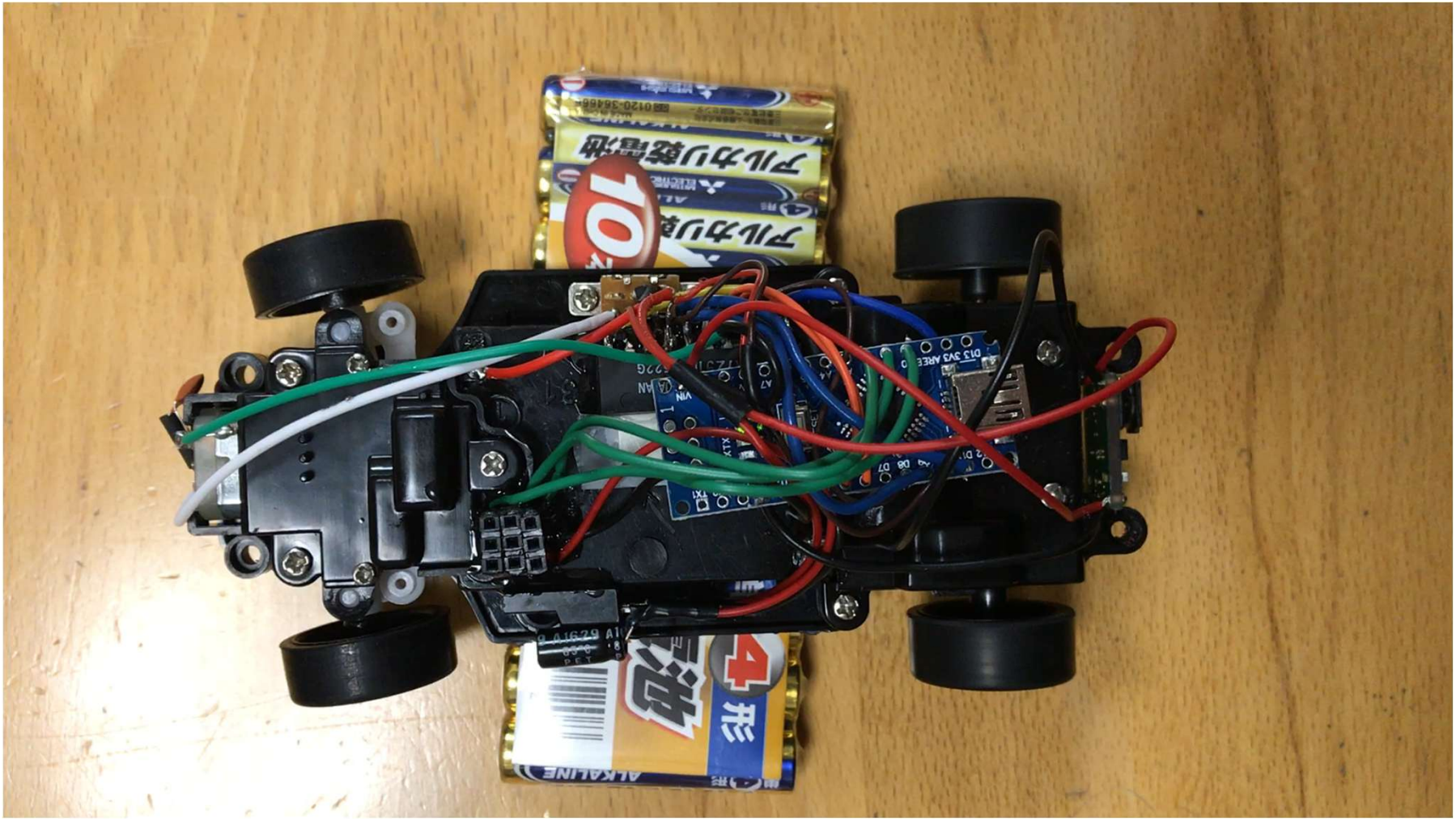
Development Photo



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Operation Test (Video)

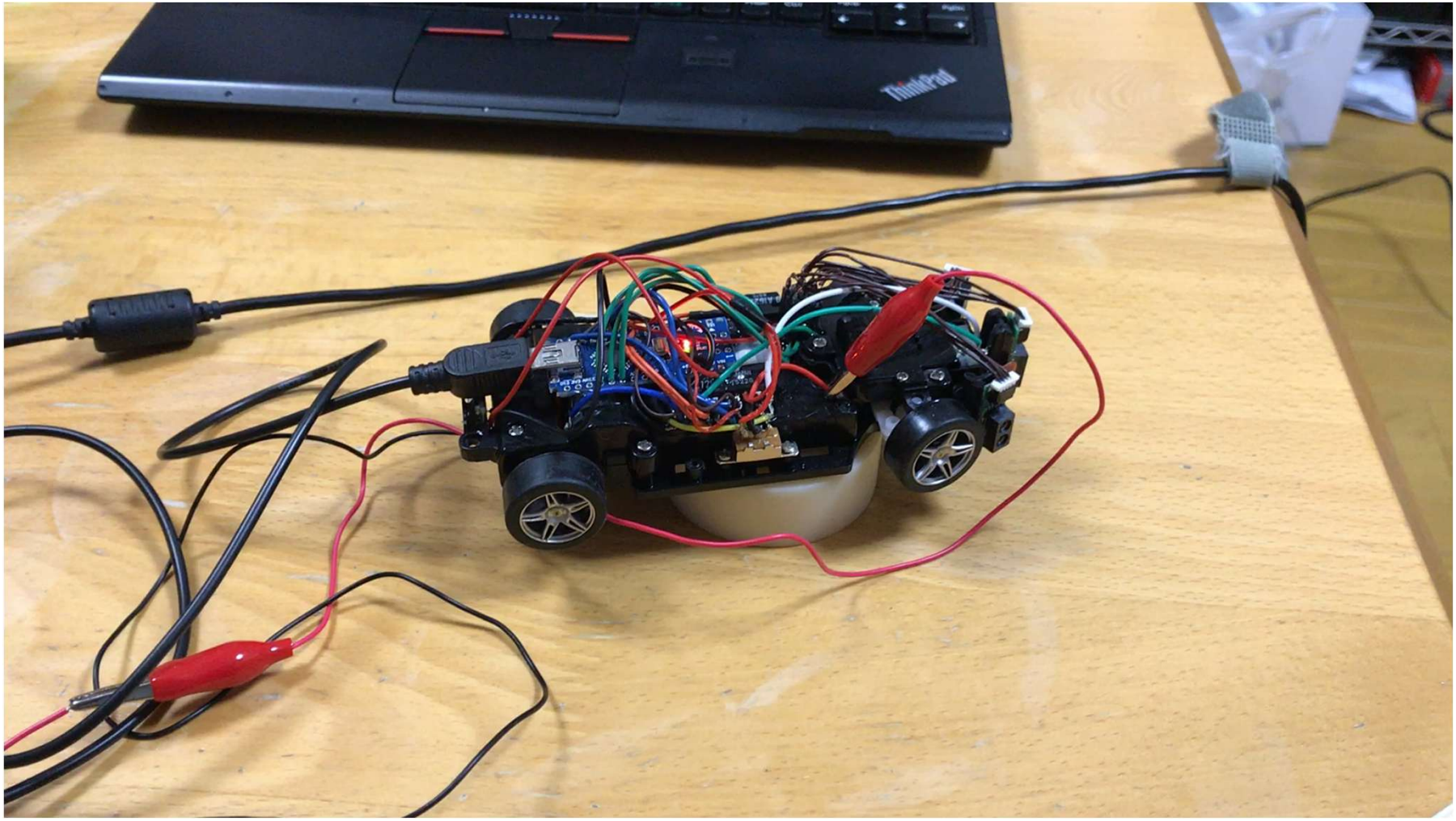
https://youtu.be/8jULg3fZ-jw



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Operation Test (Video)

https://youtu.be/8jULg3fZ-jw



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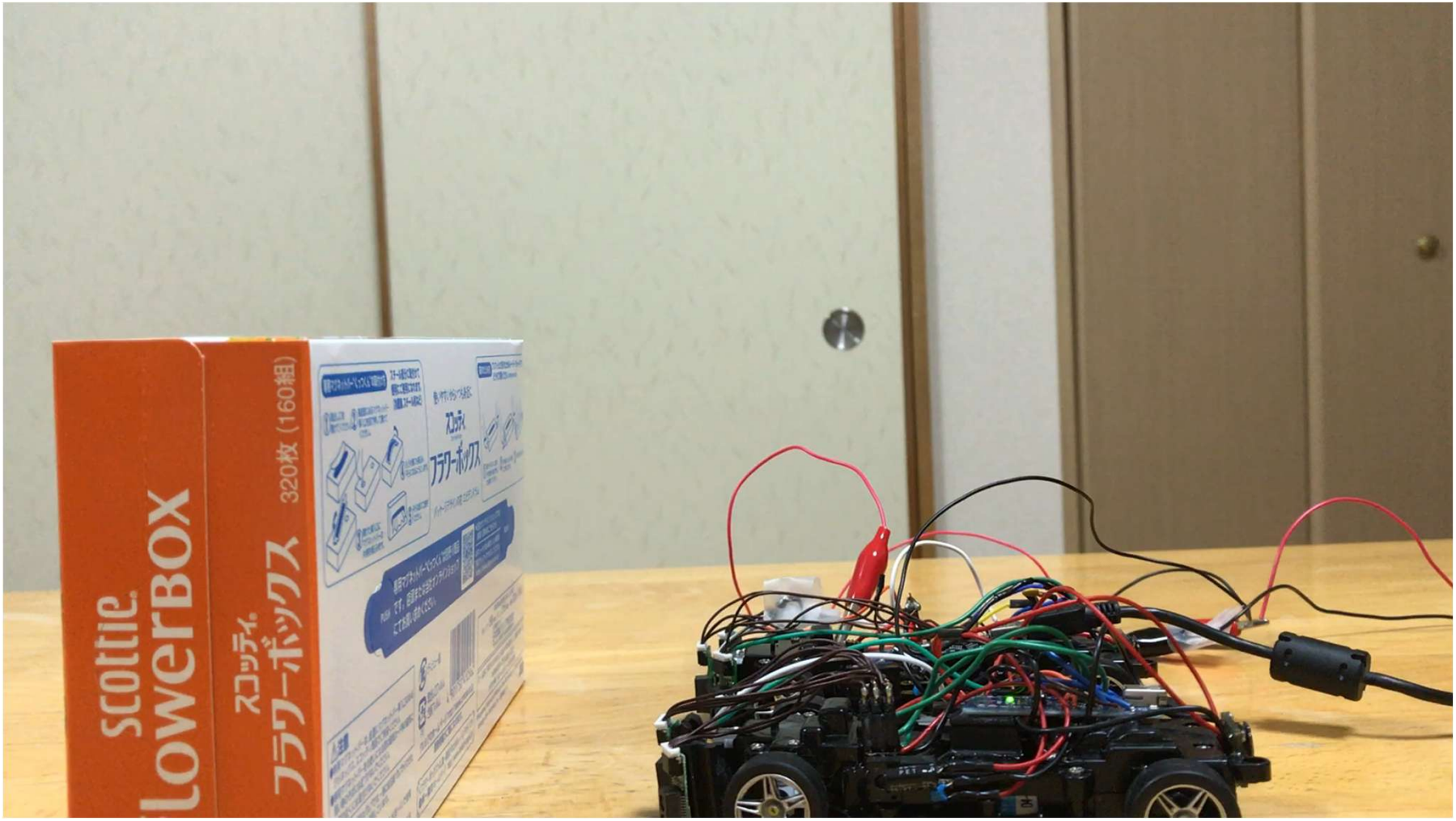
System of Car No.4



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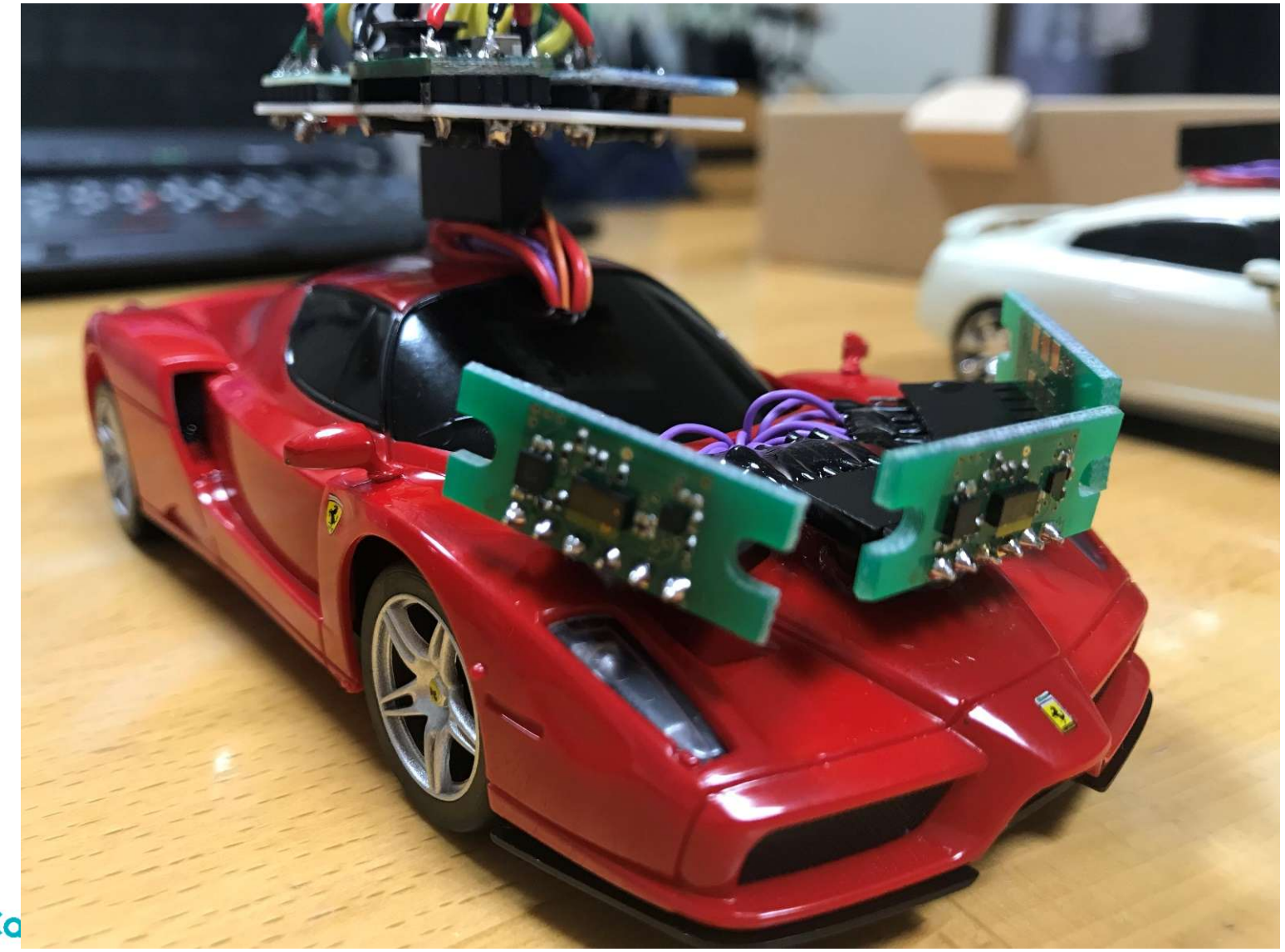
Operation Test (Video)

https://youtu.be/dP815X6oKPw

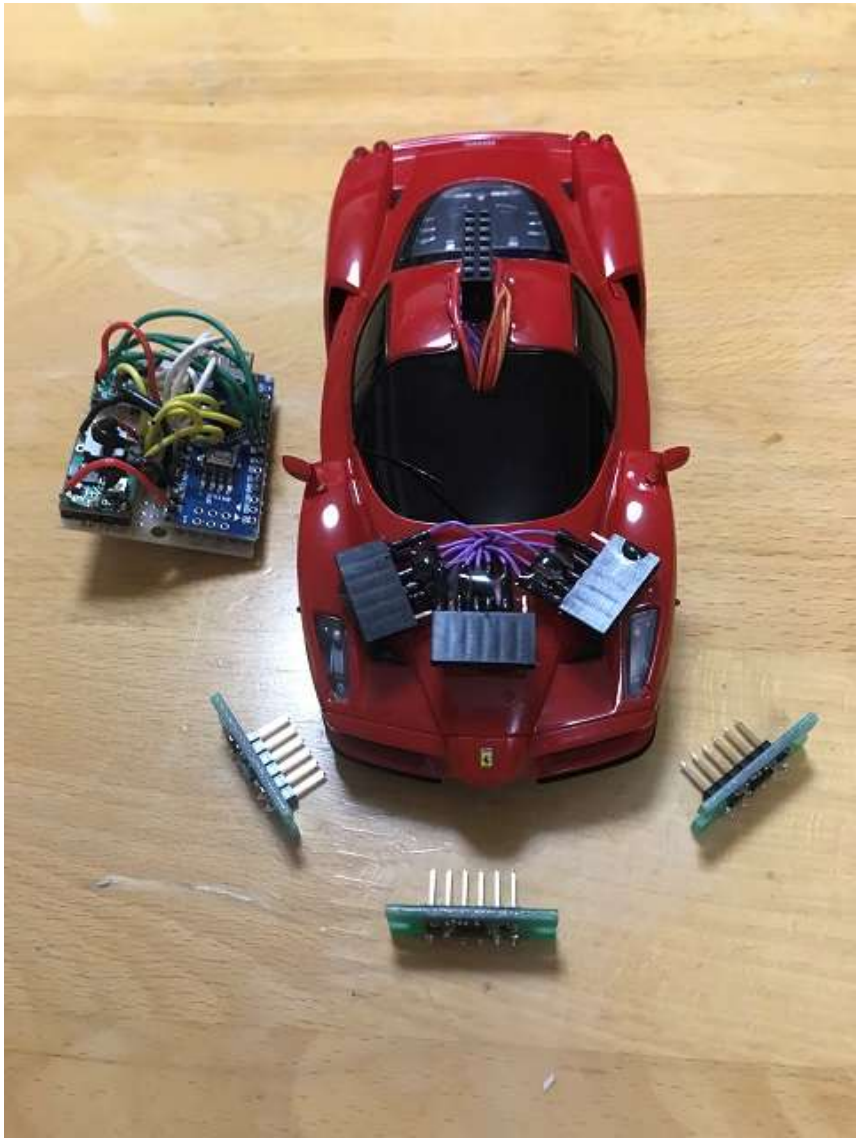


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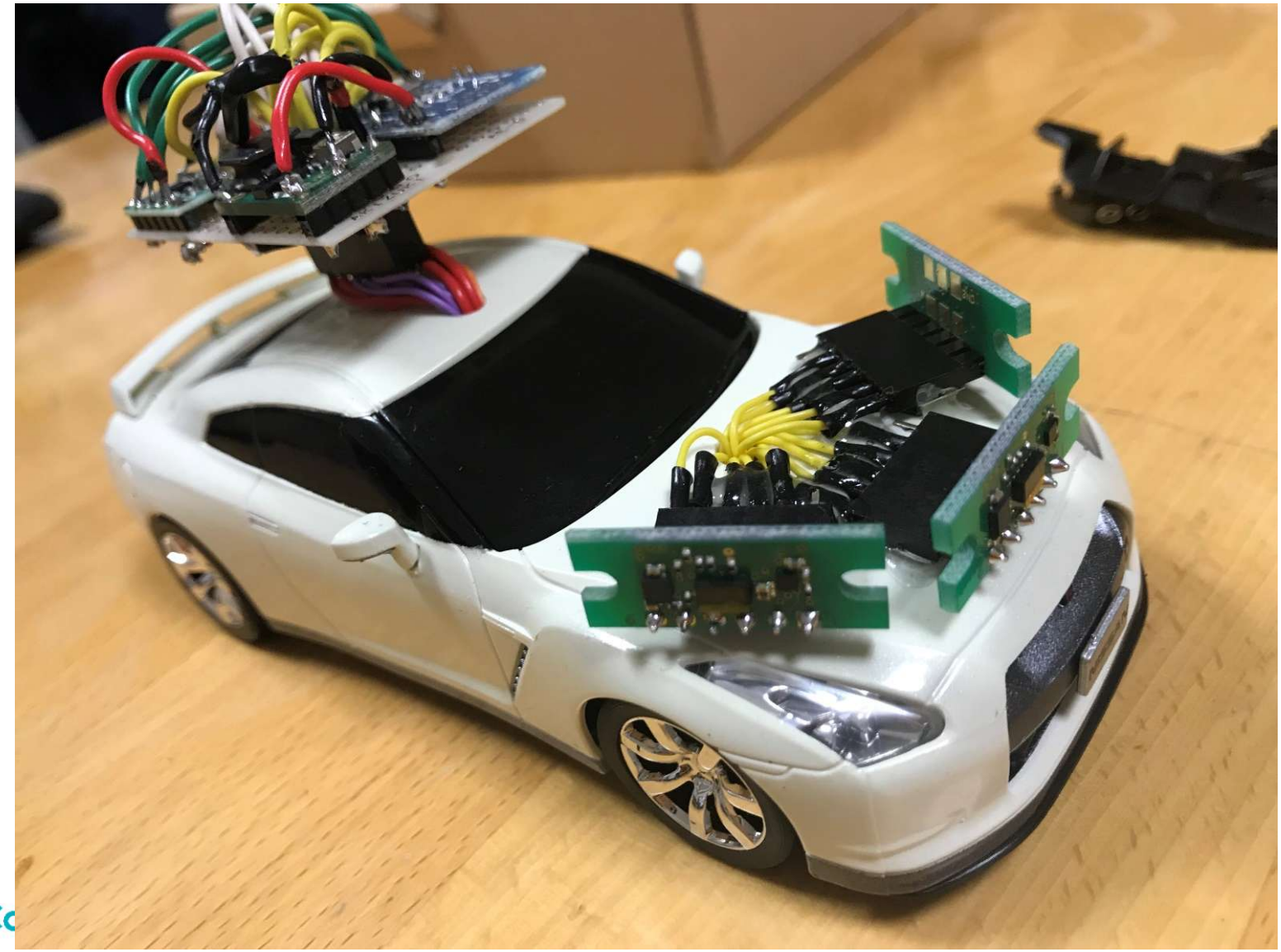
Car No.6



Car No.6



Car No.7



Standardized Interface



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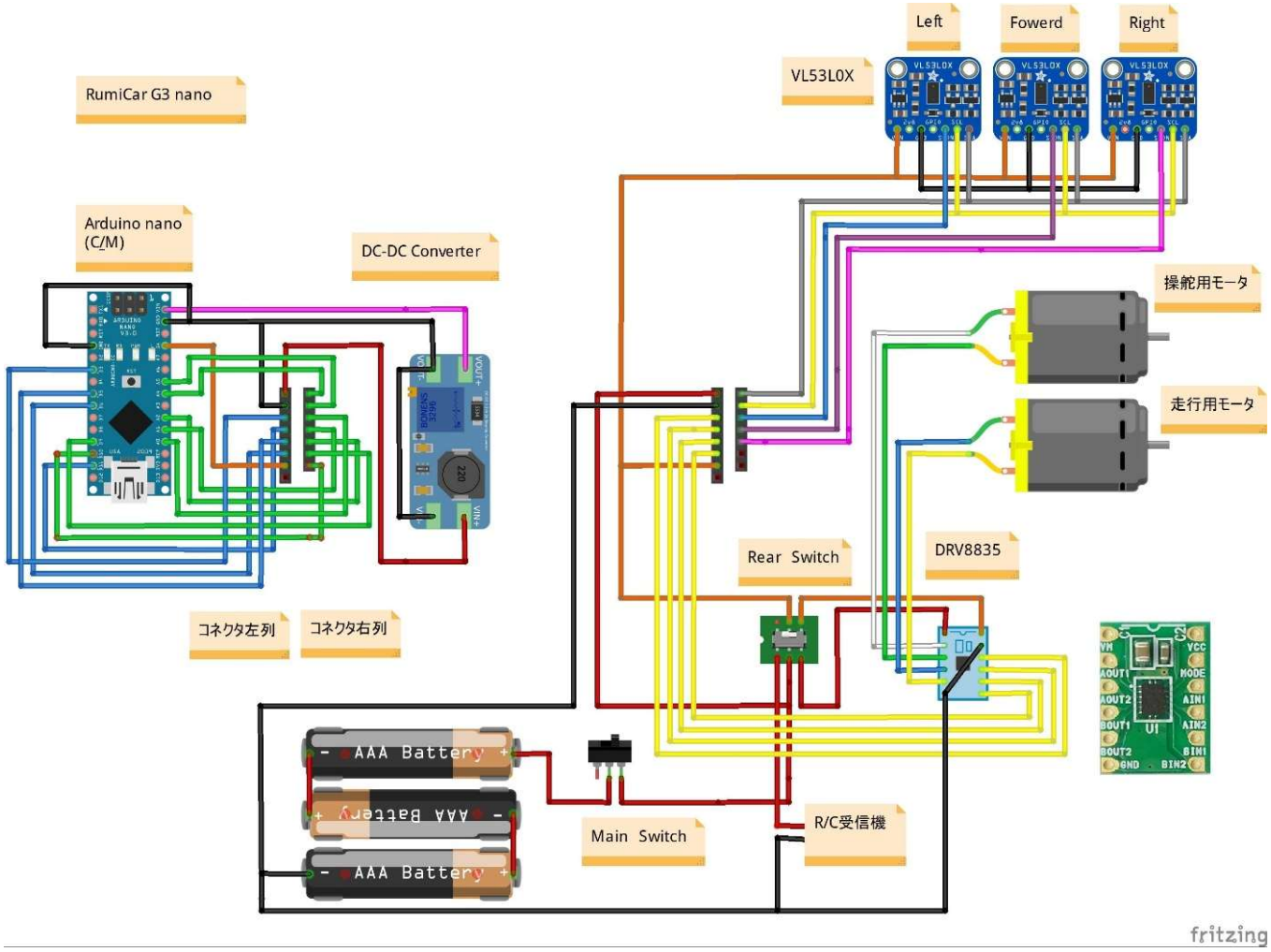
RumiCar History & Configuration

Car No.	No.1	No.2	No.3	No.4	No.5
Base car scale	1/24		1/32		
Controller	Arduino UNO		Arduino Nano		Modular (selectable / interchangeable)
Motor Control	Relay shield		MM-531	TA7291P	DRV8835
Ranging module	GP2Y0A21YK		GP2Y0E02A		VL53L0X
Ranging method	Infrared				Laser
Arduino power	Six AA batteries (9V)		Boosted from AAA batteries (4.5V) to 7V with an LMR62421		
Ranging module power	Supplied from the Arduino		3.3V generated by a TA48033S		

BD6212HFP motor driver: too small, not used

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3rd Generation Wiring Diagram



© RumiCar Development Team

Interface

Old type



NEW



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8×7 Pin Interface

表 3□第三世代□車体-CM 接続インターフェイス(EX-IF)(旧表記)

I/O	用途	ピン番号		用途	I/O
O	電源(4.5V)	1	8	SDA:E-GPIO21	I
O	GND	2	9	SCL:E-GPIO22	I
I	AIN1:右折側:A3;·E-GPIO04	3	10	SHDN0(左):E-GPIO19	I
I	AIN2:左折側:A11:·E-GPIO26	4	11	SHDN1(中):E-GPIO18	I
I	BIN1:前進側:A5:·E-GPIO27	5	12	SHDN2(右):E-GPIO05	I
I	BIN2:後進側:A6:·E-GPIO25	6	13	SERVO(操舵)用 A9 番ピン E-GPIO32	I
I	信号最大電圧及び電力供給(通常コンピュータの VCC)(5V または 3.3V)VL53L0X と DRV8835 の VCC へ	7	14	SERVO 用 A10 番ピン E-GPIO33	I
	誤挿入防止用のダミーピン (接続先は無し)	7.5			

Added 7.5 as a dummy pin to prevent incorrect insertion

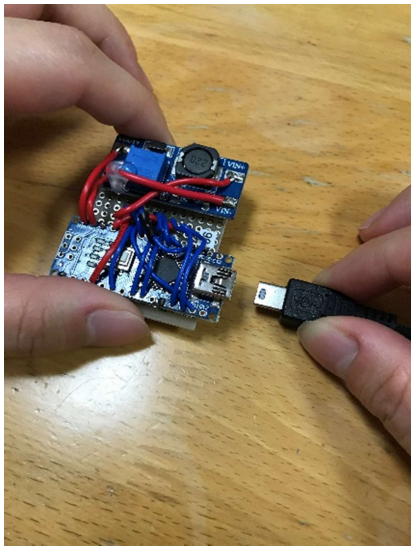
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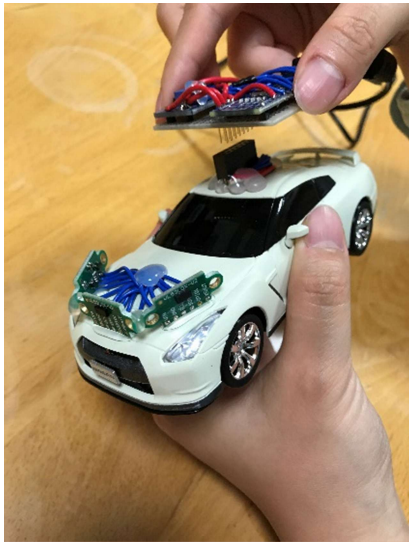
Preparation

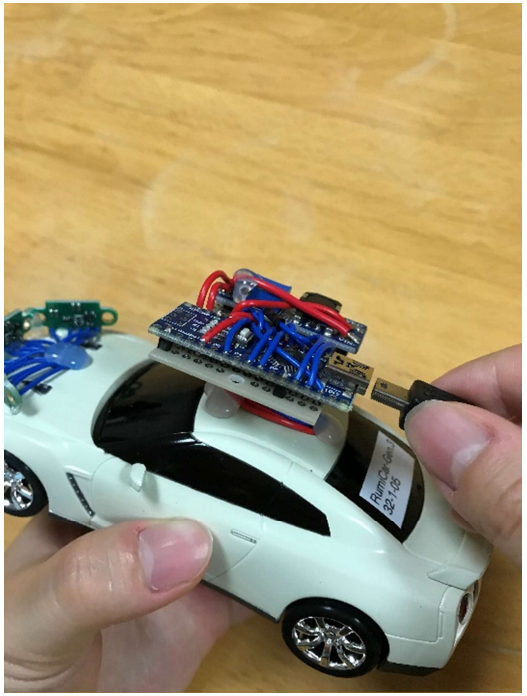
Notes, What You Need, and Preparation

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Cable Connection Cautions

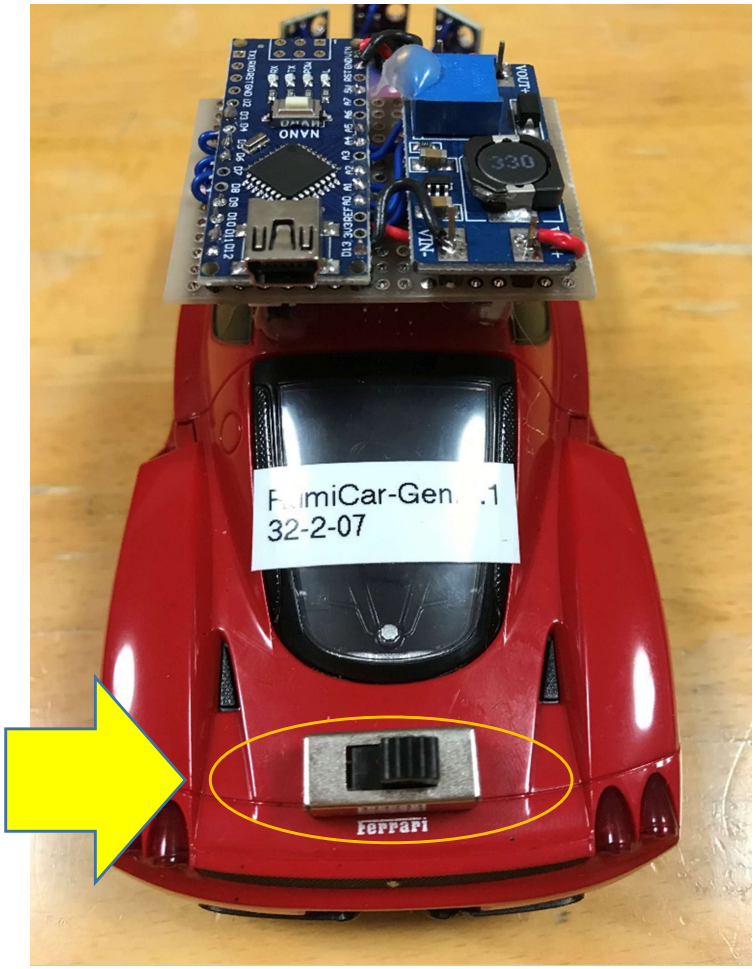






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Set to Computer Control



- (Some models only)
- On models with a switch between manual control and computer control, set the switch to the right
- The right side is computer control

- On the right, the car runs on commands from the CM (Arduino, ESP32, etc.)
- (On the left, it can be operated by radio control)

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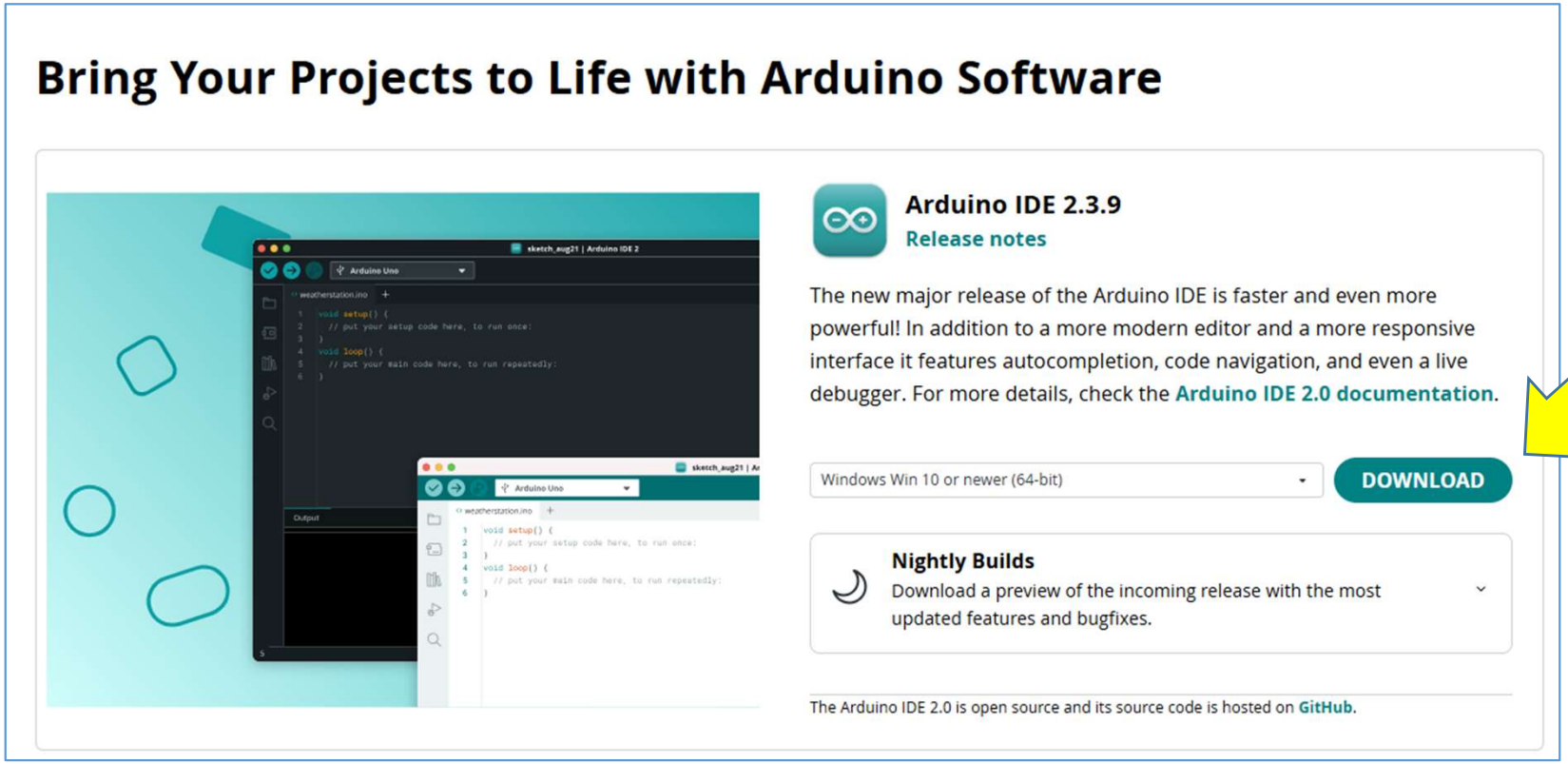
Cable Connection Cautions

- First, connect the USB cable to the compute module
- Then connect the compute module (hereafter CM) to the car body
- Caution! Do not connect the USB cable to the compute module while the CM is attached to the car body
- Do not unplug the USB cable while the CM is attached to the car body
- Excessive force on the CM connector can cause damage

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Installing the Arduino IDE

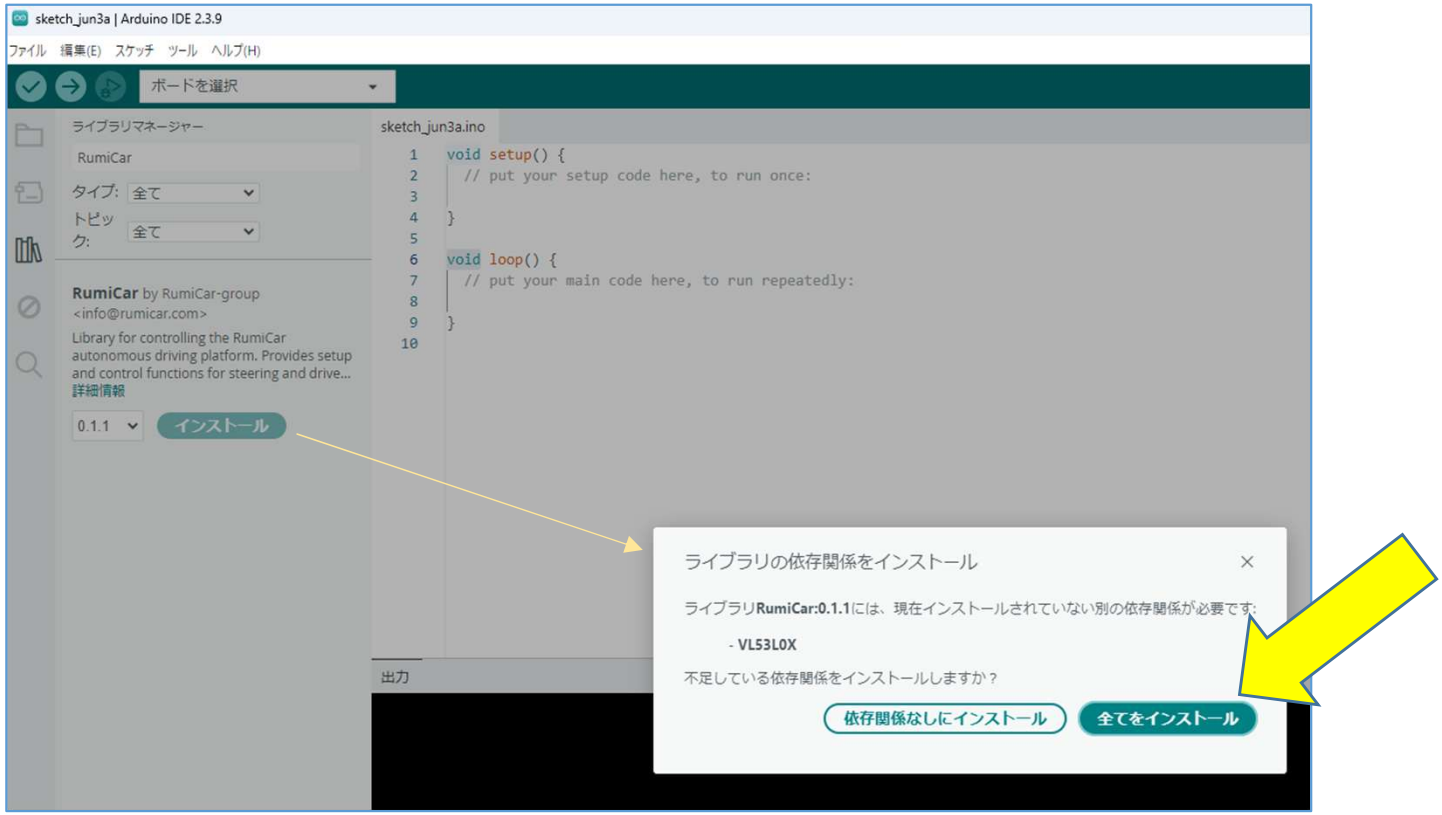
- URL:https://www.arduino.cc/en/Main/Software
- Install "Windows Win 10 or newer (64-bit)"
- (If you have already installed it, you can skip this step)



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Installing the RumiCar Library

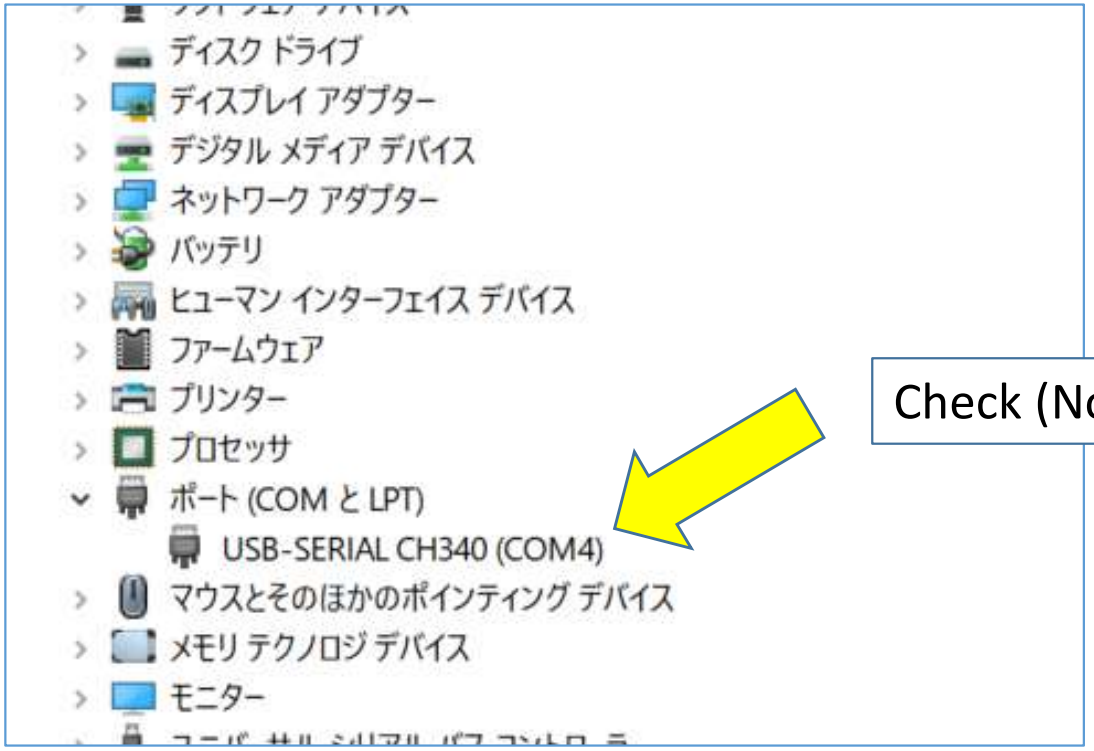
- Open the Arduino IDE
  - In "Tools" -> "Manage Libraries...", search for "RumiCar" and press Install. Choose "Install All" to include dependencies



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Checking the COM Port (Windows)

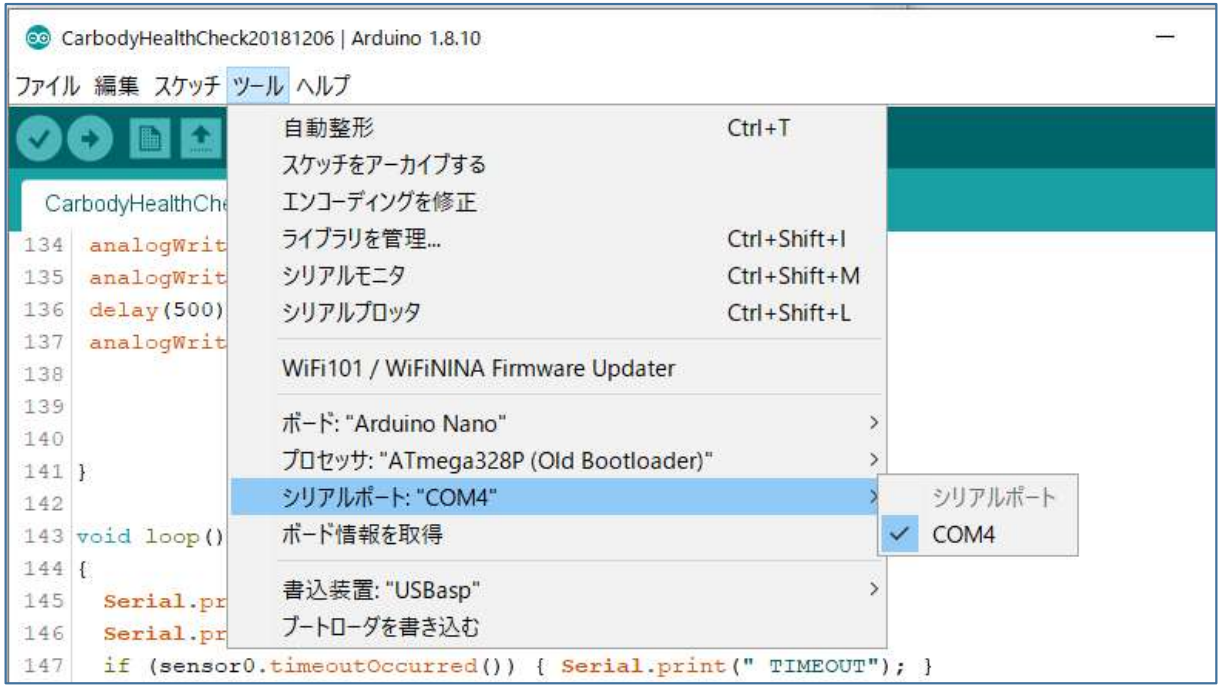
- Connect the RumiCar compute module to your computer with a USB cable
- In Device Manager, check the number of the connected COM port. Look for CH340 (remember the COM number)



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Arduino IDE Settings

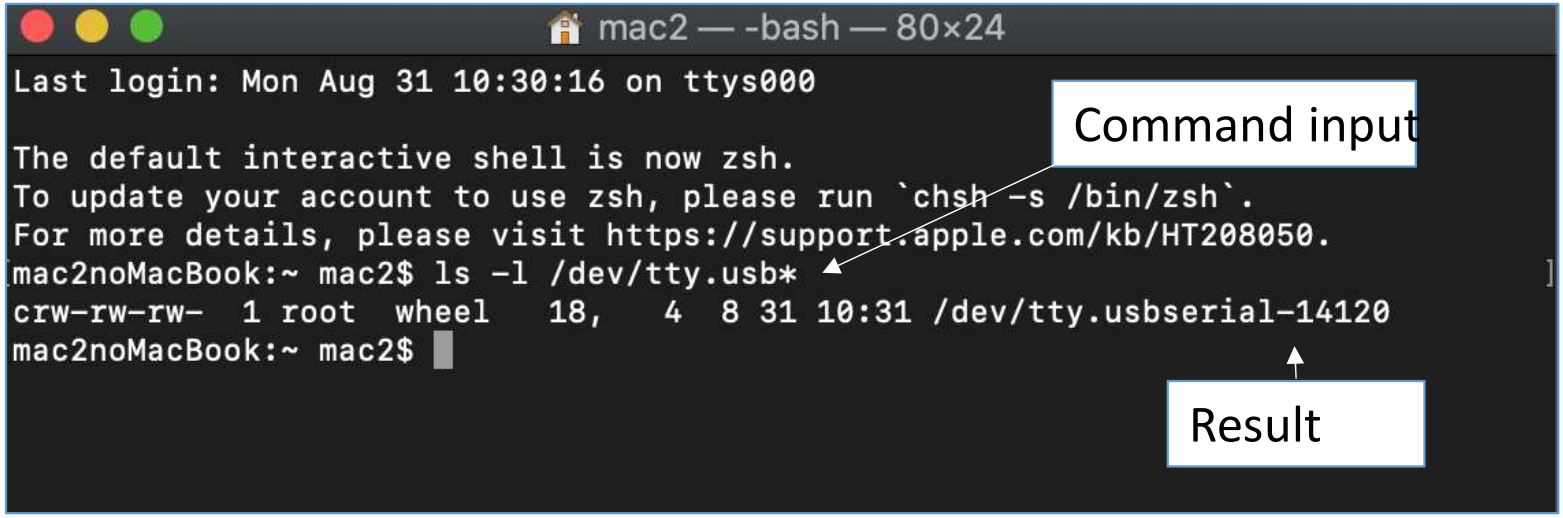
- Board: Arduino Nano
- Processor: ATmega328P (Old Bootloader) *For a CM with a Type-C port: "ATmega328P"
- Serial port: the number you checked in Device Manager



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Checking the COM Port (Mac)

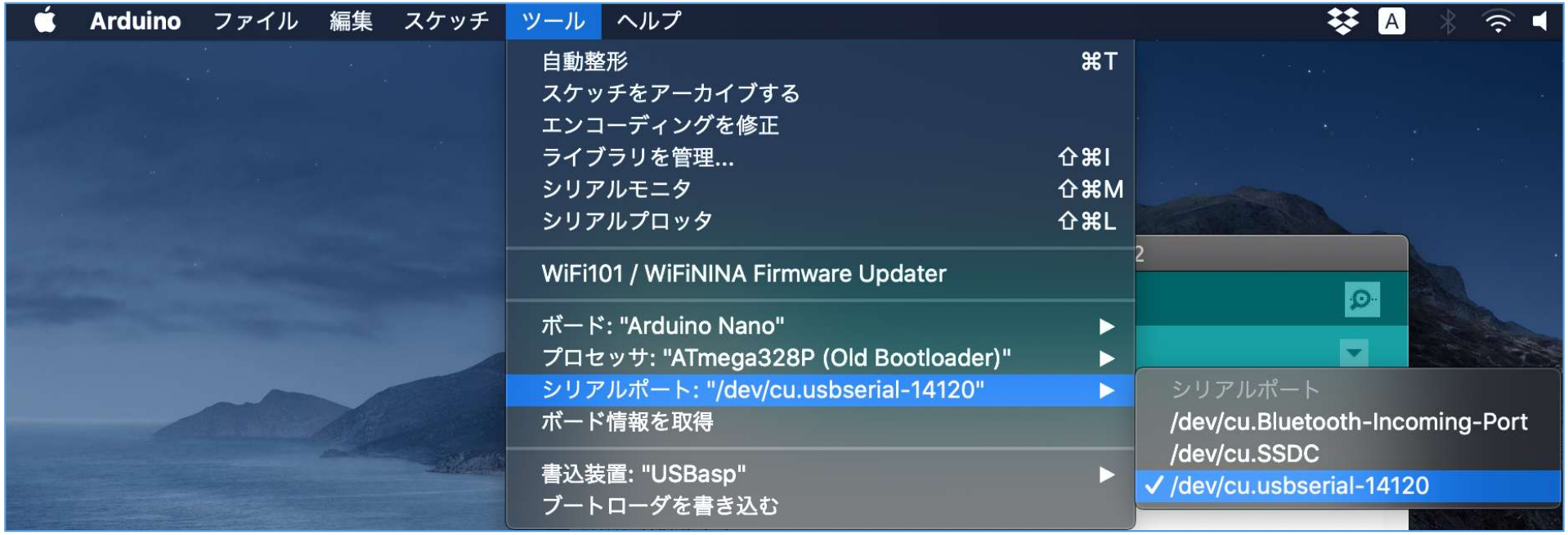
- Start the Terminal app
- Enter the command "ls –l /dev/tty.usb*" to list the connected COM ports
- Check the COM port the Arduino is connected to
- In the example below, "usbserial-14120" (remember it)



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Arduino IDE Settings (Mac)

- Board: Arduino Nano
- Processor: ATmega328P (Old Bootloader)
- Serial port: the device you checked in Terminal



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Confirm Settings

- Important! Check once more!

Item	Setting	Notes
Board	Arduino Nano	Does not work with Arduino UNO
Processor	ATmega328P(Old Bootloader)	Old Bootloader version (Type-C CM: "ATmega328P")
Serial port	"The number you checked in Device Manager"	May change if you change boards
Programmer	(No change needed)	Not used for normal uploading
Serial port baud rate	9600bps	Must be 9600bps (Arduino Nano) Check in the Serial Monitor

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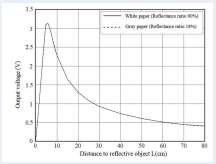
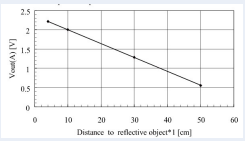
Distance Measurement

Laser Ranging Module

© RumiCar Development Team

Ranging Modules

- Comparison of Ranging Modules

Model number	GP2Y0A21YK0F	GP2Y0E02A	VL53L0X	HC-SR04
Measurement method	Infrared	Infrared	Laser	Ultrasonic
Output			Digital value	Analog
Recommended supply voltage	5V	3.3V	2.6-5.5V (Board dependent)	5V
Power consumption	40mA max.	36mA max.	20mA	15mA
Measurable distance	80cm	50cm	2m	2m
Used in RumiCar	Car No.1, 2	Car No.3, 4	Car No.5 and later	Not used
Notes	Output not proportional to distance			Echo issues at high speed

Graphs from Sharp datasheets

© RumiCar Development Team

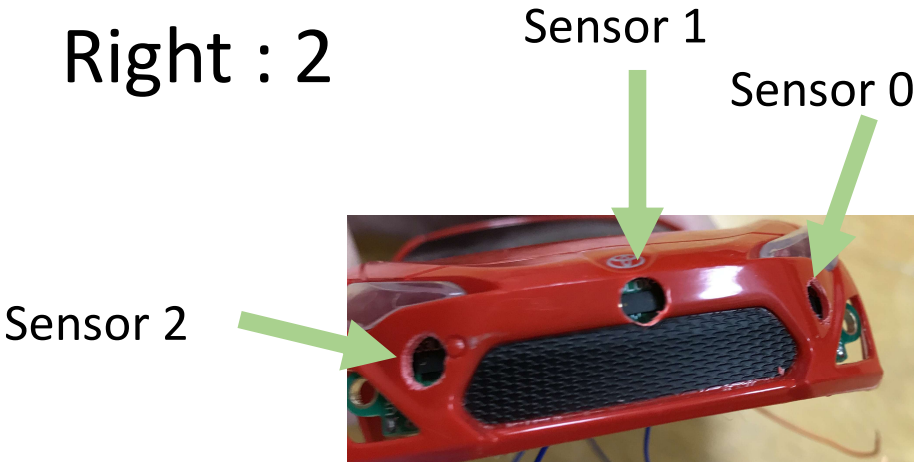
Laser Ranging Module

- 3 modules on the front of the car

Left : 0

Center: 1

Right : 2





© RumiCar Development Team

VL53L0X

- The laser ranging module on the RumiCar is the VL53L0X
- Features of the VL53L0X
  - Can measure up to 2m
  - Measurements can be read via I2C
  - I2C address can be reassigned by software
  - Modes can be roughly set with parameters
    - High-speed mode
    - Long-range mode
    - High-accuracy mode
    - The sample programs are set to high-speed mode

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Ranging Methods

- To get the measurement from each ranging module, use for example the command below. The result is an integer (int) in mm (millimeters)

  readRangeSingleMillimeters()

- For example, to get the measurement of sensor 1:

  i = sensor1.readRangeSingleMillimeters();

- (Reference) You can get the same value with the library function RC_read

  i = RC_read(CENTER);

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Opening Example Sketches

(No need to download from GitHub)

From the Arduino IDE menu

"File"

▼

"Examples"

▼

"RumiCar" (near the bottom of the list)

▼

"Exercise-1.1", etc.

*Requires the RumiCar library (previous step)

[Ref.] Advanced programs (A2.3, etc.) are on GitHub (not used today)



Exercise headers, refactoring and advanced programs added by Mr. Shigeki INADAMA

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Exercise-1

Let's Measure Distance

© RumiCar Development Team

Exercise-1.1

Ranging with the Central Sensor!

© RumiCar Development Team

Exercise-1.1
Central Sensor Ranging (Prep)

- Turn the RumiCar's power off
- Remove the CM from the RumiCar body
- Connect the USB cable to the CM
- Connect the USB cable to your computer
- Attach the CM to the RumiCar body

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Exercise-1.1
Central Sensor Ranging

- Open "Exercise" from the Examples
- Add one line of code in loop() to display the measurement

```
 8  void loop()
 9  {
10
11  }
```



```
 8  void loop()
 9  {
10    Serial.println(sensor1.readRangeSingleMillimeters());
11  }
```

- The Serial.print and Serial.println commands display data on the Serial Monitor
- Serial.print: no line break
- Serial.println: line break after printing

© RumiCar Development Team

Exercise-1.1
Central Sensor Ranging

- Compile and Upload to the Board

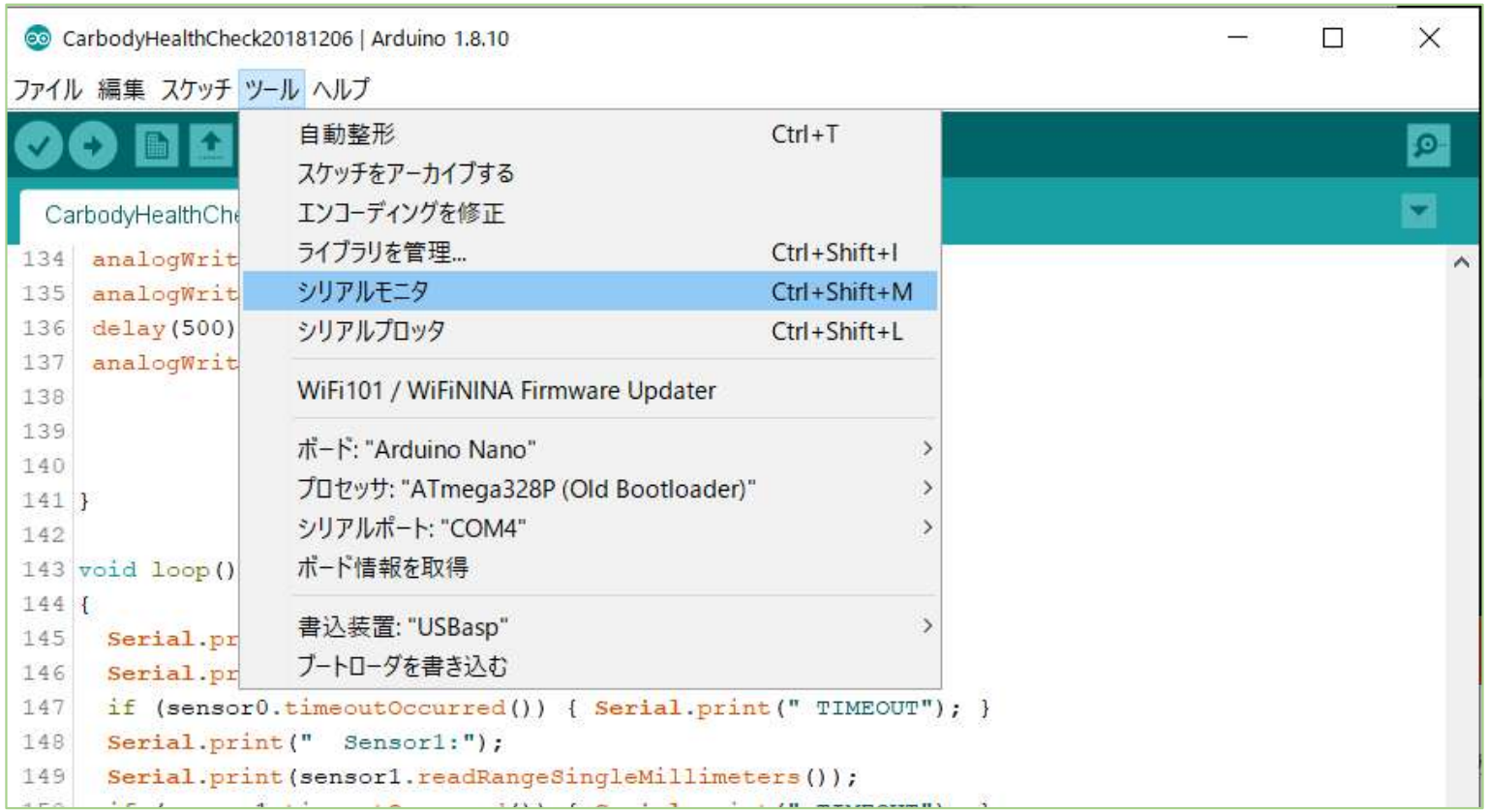
Click



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Exercise-1.1
Central Sensor Ranging (Cont'd)

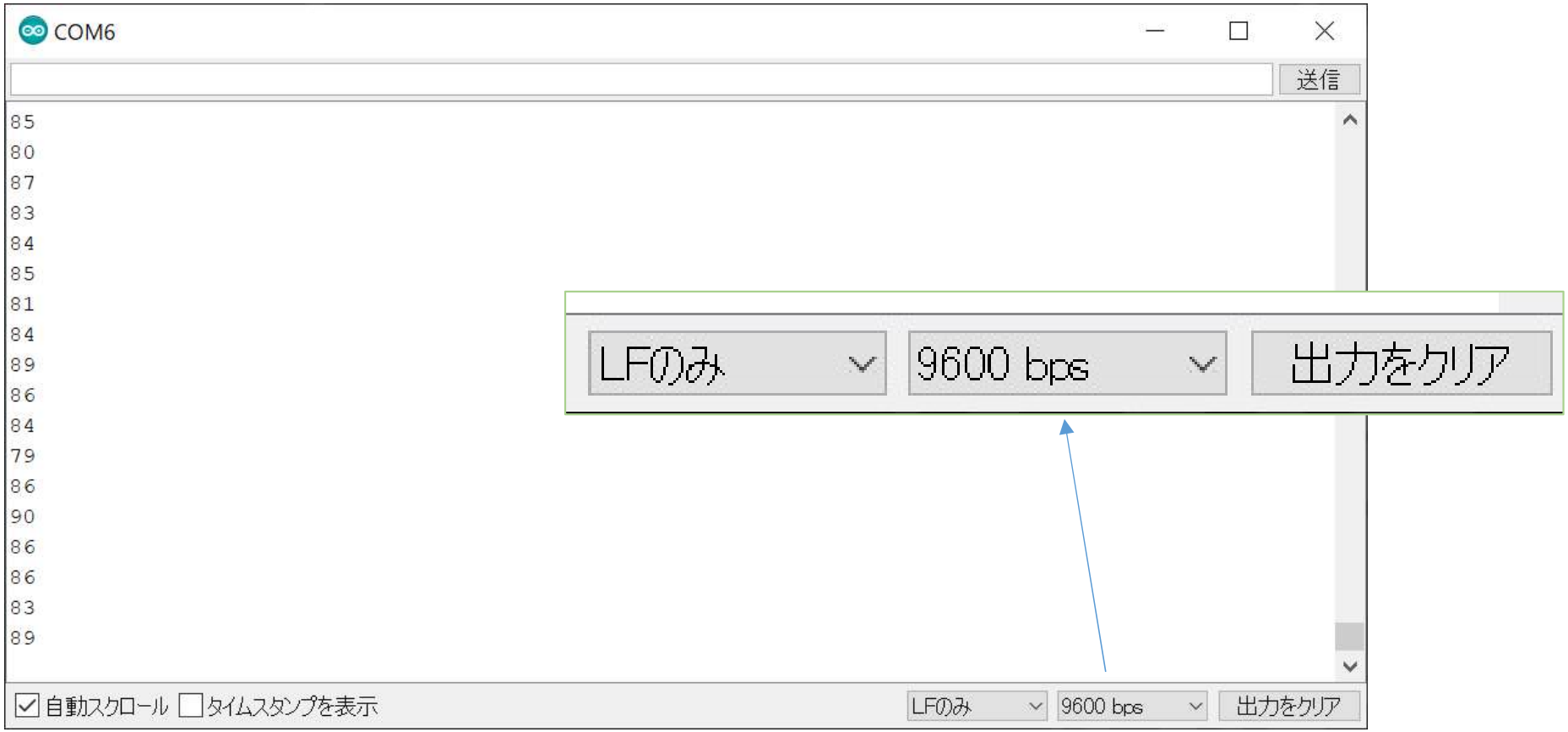
- Display the measurements on the Serial Monitor
  - Tools -> Serial Monitor



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Exercise-1.1
Central Sensor Ranging (Cont'd)

- If the display does not work, check the baud rate (it must be 9600 bps)



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Exercise-1.1
Central Sensor Ranging (Cont'd)

- If the display still does not work, check the "Confirm Settings" page in the "Preparation" section again

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Exercise-1.2

Ranging with 3 Sensors!

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Exercise-1.2
Ranging with 3 Sensors

- Open "Exercise-1.2" from the Examples

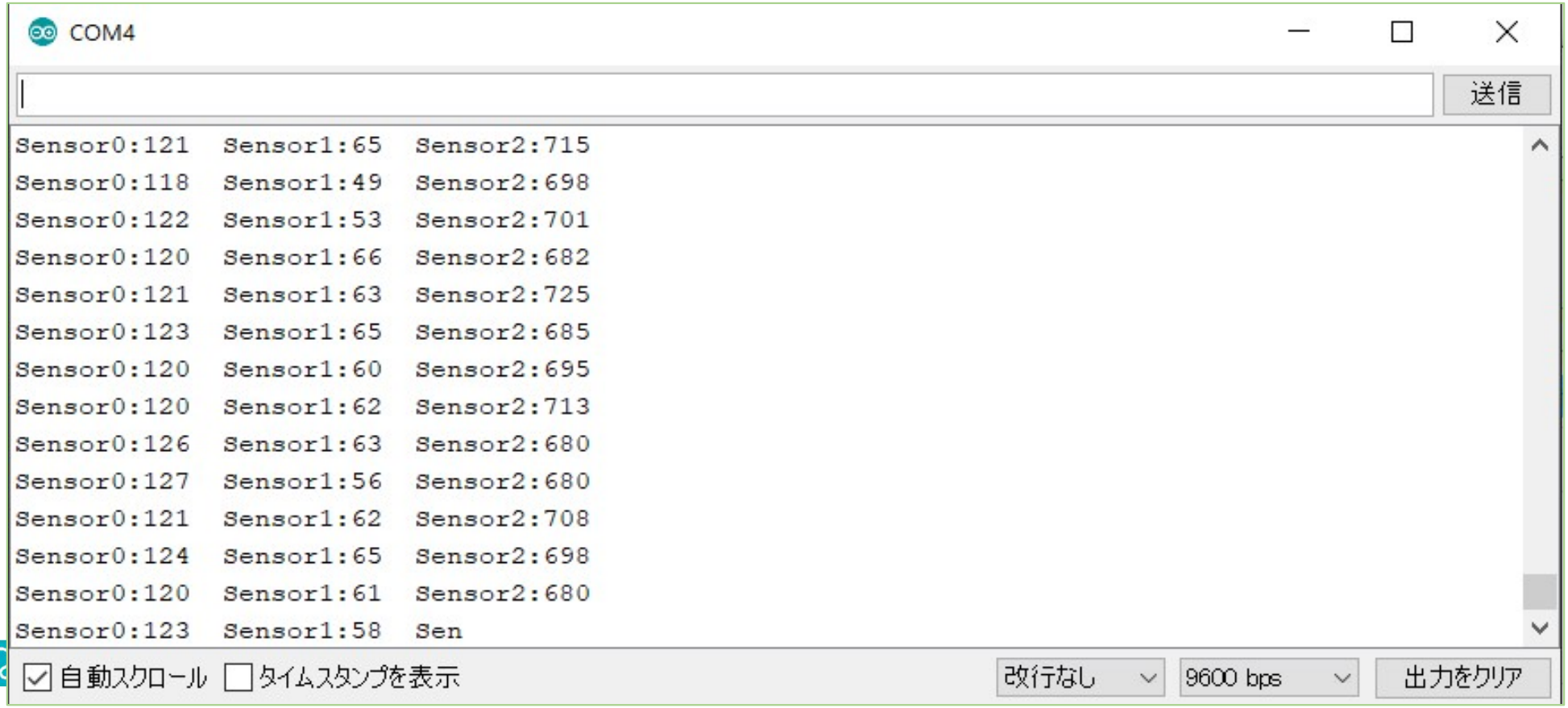
Sensors	Sensor location
Sensor0	Left
Sensor1	Center
Sensor2	Right

```
 8  void loop()
 9  {
10    Serial.print("Sensor0:");
11    Serial.print(sensor0.readRangeSingleMillimeters());
12    Serial.print("  Sensor1:");
13    Serial.print(sensor1.readRangeSingleMillimeters());
14    Serial.print("  Sensor2:");
15    Serial.println(sensor2.readRangeSingleMillimeters());
16  }
```


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Exercise-1.2
Ranging with 3 Sensors (Cont'd)

- Display the measurements on the Serial Monitor
  - Tools -> Serial Monitor



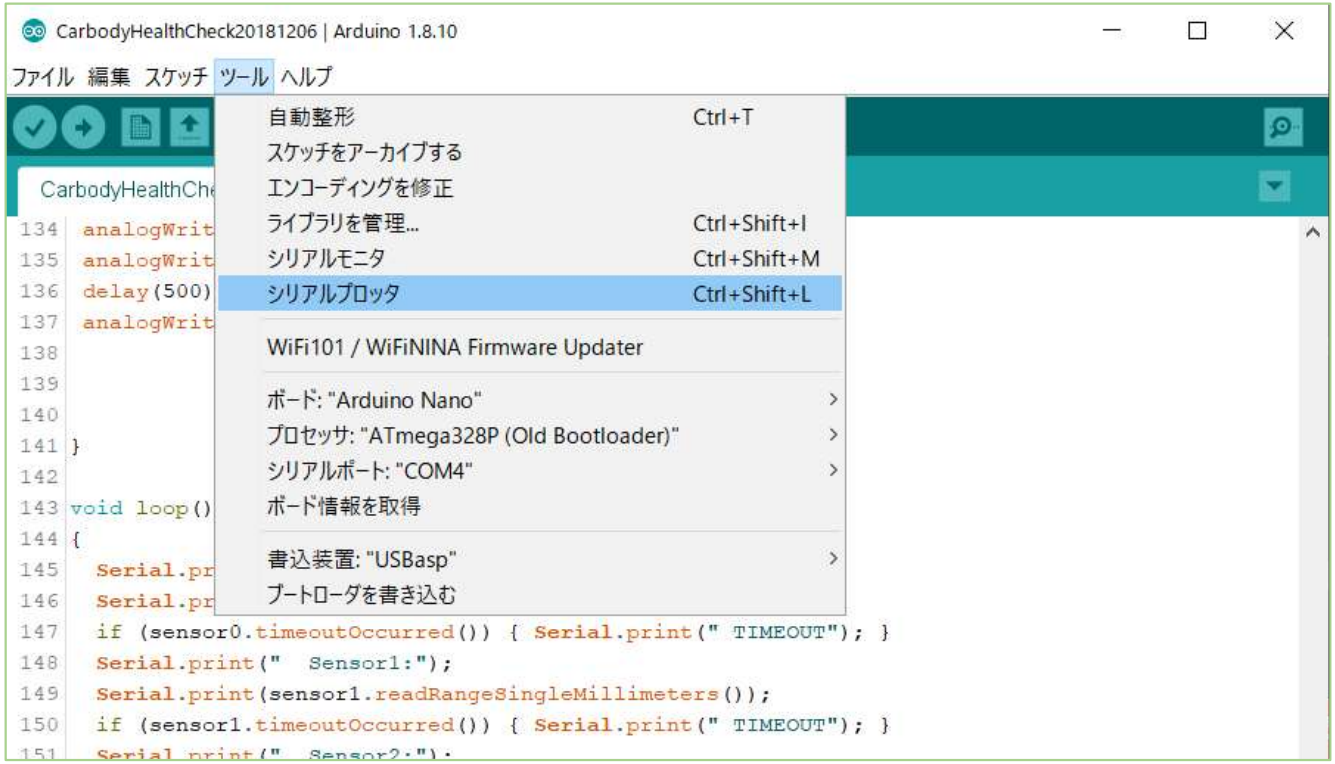
Exercise-1.3

Using the Serial Plotter

© RumiCar Development Team

Exercise-1.3
Using the Serial Plotter

- (Use the program from Exercise-1.2 as is)
- Close the Serial Monitor
- Tools -> Serial Plotter
  - *If the Serial Monitor window is open, you must close it

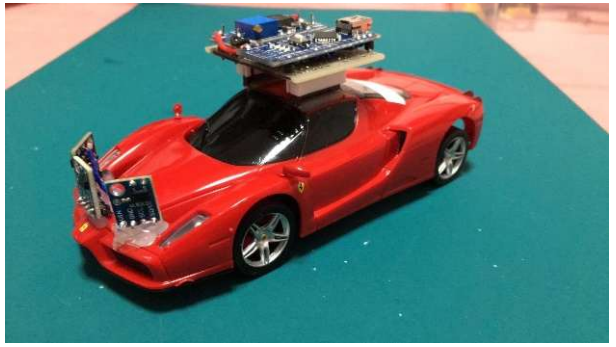


Break Room



	UV-curing resin	Hot-melt adhesive (commonly called glue)
Pros	• Hardens quickly • High strength • Transparent, looks good • Can be applied in layers	• Cheap • Easy to peel off and redo
Cons	• Brittle over a long time	• Not transparent; looks worse (not Insta-worthy) • Weaker than UV resin





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Motor Control

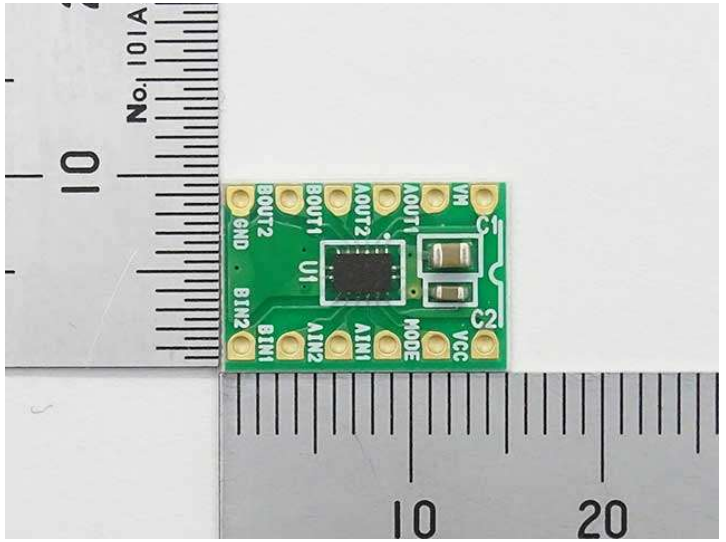
Moving the Steering and Driving

© RumiCar Development Team

Motor Driver

- The motor driver on the RumiCar is the DRV8835
- Features of the DRV8835
  - 2 channels
  - Up to 1.5A per channel
  - PWM(250kHz)
  - Low power consumption
  - Low voltage drop (motor side)
  - Logic power (VCC) and motor power are independent
    - Logic supply voltage: 2 to 7V
    - Motor supply voltage: up to 11V

DRV8835



© RumiCar Development Team

DRV8835 Specs (1/2)

T_A = 25°C (unless otherwise noted)

		MIN	NOM	MAX	UNIT
V_{CC}	Device power supply voltage	2		7	V
V_M	Motor power supply voltage	0		11	V
V_{IN}	Logic level input voltage	0		V_{CC}	V
I_{OUT}	H-bridge output current(1)	0		1.5	A
f_{PWM}	Externally applied PWM frequency	0		250	kHz

(1) Power dissipation and thermal limits must be observed.

From the DRV8835 Dual Low-Voltage H-Bridge IC datasheet, August 9, 2016

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DRV8835 Specs (2/2)

T_A = 25°C, V_M = 5 V, V_{CC} = 3 V (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
POWER SUPPLY						
I_{VM}	VM operating supply current	No PWM, no load		85	200	µA
		50 kHz PWM, no load		650	2000	
I_{VMQ}	VM sleep mode supply current	V_M = 2 V, V_{CC} = 0 V, all inputs 0 V		5		nA
		V_M = 5 V, V_{CC} = 0 V, all inputs 0 V		10	95	
I_{VCC}	VCC operating supply current			450	2000	µA
V_{UVLO}	VCC undervoltage lockout voltage	V_{CC} rising			2	V
		V_{CC} falling			1.9	
LOGIC-LEVEL INPUTS						
V_{IL}	Input low voltage				0.3 × V_{CC}	V
V_{IH}	Input high voltage		0.5 × V_{CC}			V
I_{IL}	Input low current	V_{IN} = 0	–5		5	µA
I_{IH}	Input high current	V_{IN} = 3.3 V			50	µA
R_{PD}	Pulldown resistance			100		kΩ
H-BRIDGE FETS						
$R_{DS(ON)}$	HS + LS FET on resistance	V_{CC} = 3 V, V_M = 3 V, I_O = 800 mA, T_J = 25°C		370	420	mΩ
		V_{CC} = 5 V, V_M = 5 V, I_O = 800 mA, T_J = 25°C		305	355	
I_{OFF}	OFF-state leakage current				±200	nA
PROTECTION CIRCUITS						
I_{OCP}	Overcurrent protection trip level		1.6		3.5	A
t_{DEG}	Overcurrent de-glitch time			1		µs
t_{OCR}	Overcurrent protection retry time			1		ms
t_{DEAD}	Output dead time			100		ns
t_{TSD}	Thermal shutdown temperature	Die temperature	150	160	180	°C

RumiCar

From the DRV8835 Dual Low-Voltage H-Bridge IC datasheet, August 9, 2016

IN/IN Mode

- It has an IN/IN mode and a PHASE/ENABLE mode
- RumiCar uses the DRV8835 in IN/IN mode

Table 3. IN/IN Mode

MODE	xIN1	xIN2	xOUT1	xOUT2	FUNCTION (DC MOTOR)
0	0	0	Z	Z	Coast
0	0	1	L	H	Reverse
0	1	0	H	L	Forward
0	1	1	L	L	Brake

From the DRV8835 Dual Low-Voltage H-Bridge IC datasheet, August 9, 2016

© RumiCar Development Team

Phase/Enable Mode

Table 4. Phase/Enable Mode

MODE	xENABLE	xPHASE	xOUT1	xOUT2	FUNCTION (DC MOTOR)
1	0	X	L	L	Brake
1	1	1	L	H	Reverse
1	1	0	H	L	Forward

- When Enable is 1, the motor turns
- When Enable is 0, the motor stops
- When Phase is 1, reverse direction
- When Phase is 0, forward direction

Enable sets speed, Phase sets direction

From the DRV8835 Dual Low-Voltage H-Bridge IC datasheet, August 9, 2016

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About PWM

- RumiCar uses PWM to control driving speed
- PWM stands for "Pulse Width Modulation", a technique that changes the apparent voltage by repeatedly switching the voltage on and off at short intervals

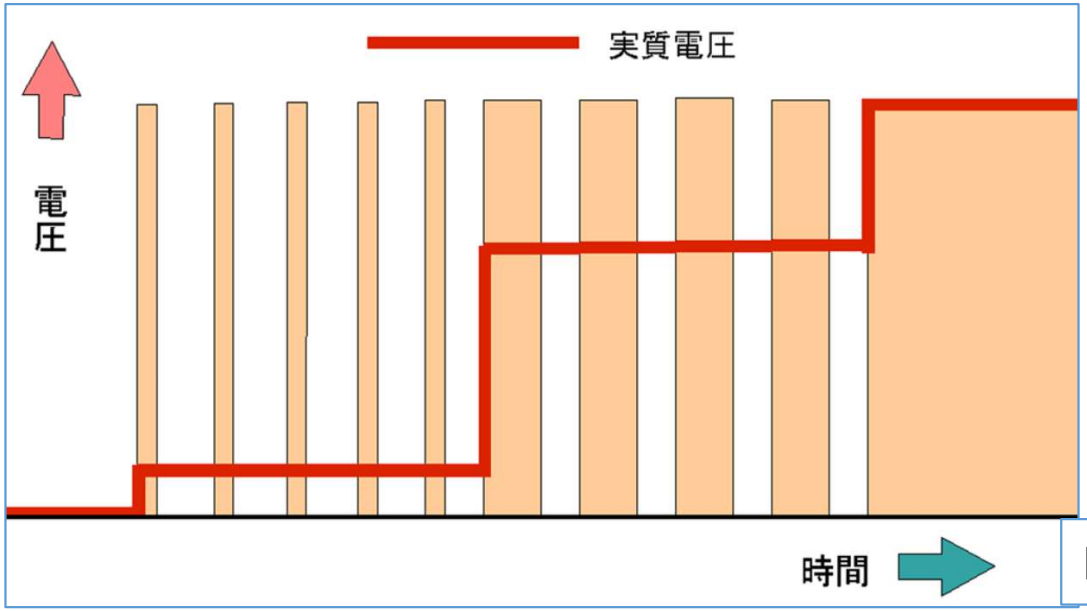


Figure from Wikipedia

PWM on Arduino

- Arduino PWM is linked to its timers
- The Arduino has 3 timers

Timer	Pin	Frequency
Timer 0	5, 6	980Hz
Timer 1	9, 10	490Hz
Timer 2	3, 11	490Hz

- Using the Arduino Servo library makes Timer 1 unavailable

Pin Assignment on the Arduino Version		
Function	Timer	Pin
Steer	Timer 2	3, 11
Drive	Timer 0	5, 6

RumiCar PWM Control

- The RumiCar sample programs define macros for the pins
- For example, to turn the steering right, set AIN1 on and AIN2 to "0"
- *Arduino pin assignment shown. ESP and other versions differ, but macros hide the differences

Timer	Function	Pin	Macro
Timer 2	Turn right	3	AIN1
	Turn left	11	AIN2
Timer 0	Forward	5	BIN1
	Reverse	6	BIN2

RumiCar RC Functions

- RumiCar provides the following functions to make program development easier

Action	Type	Function	1st argument (int)	2nd argument (int)	Action	Return value on error
Steer	int	RC_steer	RIGHT	(None)	Steer right	0
			LEFT		Steer left	
			CENTER		Center	
Drive	int	RC_drive	FREE	0 to 255 (PWM value)	Neutral	0
			REVERSE		Reverse	
			FORWARD		Forward	
			BRAKE		Brake	

- With RC_drive FREE and BRAKE, the 2nd argument must be given, but its value is ignored
- The 2nd argument of RC_drive is the PWM value; the larger the value, the more power goes to the motor
- The 1st argument is of type int. The values in this table are ultimately expanded by macros

RumiCar RC Functions (Ranging)

- A distance-measuring function, RC_read, is also provided

Action	Type	Function	1st argument (int)	Sensor to read	Return value (normal)	Return value on error
Range	int	RC_read	LEFT	Left (Sensor0)	Distance (mm) 0 to 2000	-1: timeout -2: argument error -3: out of range
			CENTER	Center (Sensor1)		
			RIGHT	Right (Sensor2)		
			REAR	Rear (Sensor3)		

- Negative values are errors: -1 no response (also when REAR is specified on a car without a rear sensor), -2 invalid argument, -3 out of range / weak signal (when readRangeSingleMillimeters() gives around 8190)
- Example: i = RC_read(CENTER); returns the same distance as sensor1.readRangeSingleMillimeters() (the Exercises use the latter)
- The rear sensor (Sensor3) is only on equipped cars. The library detects it automatically at startup

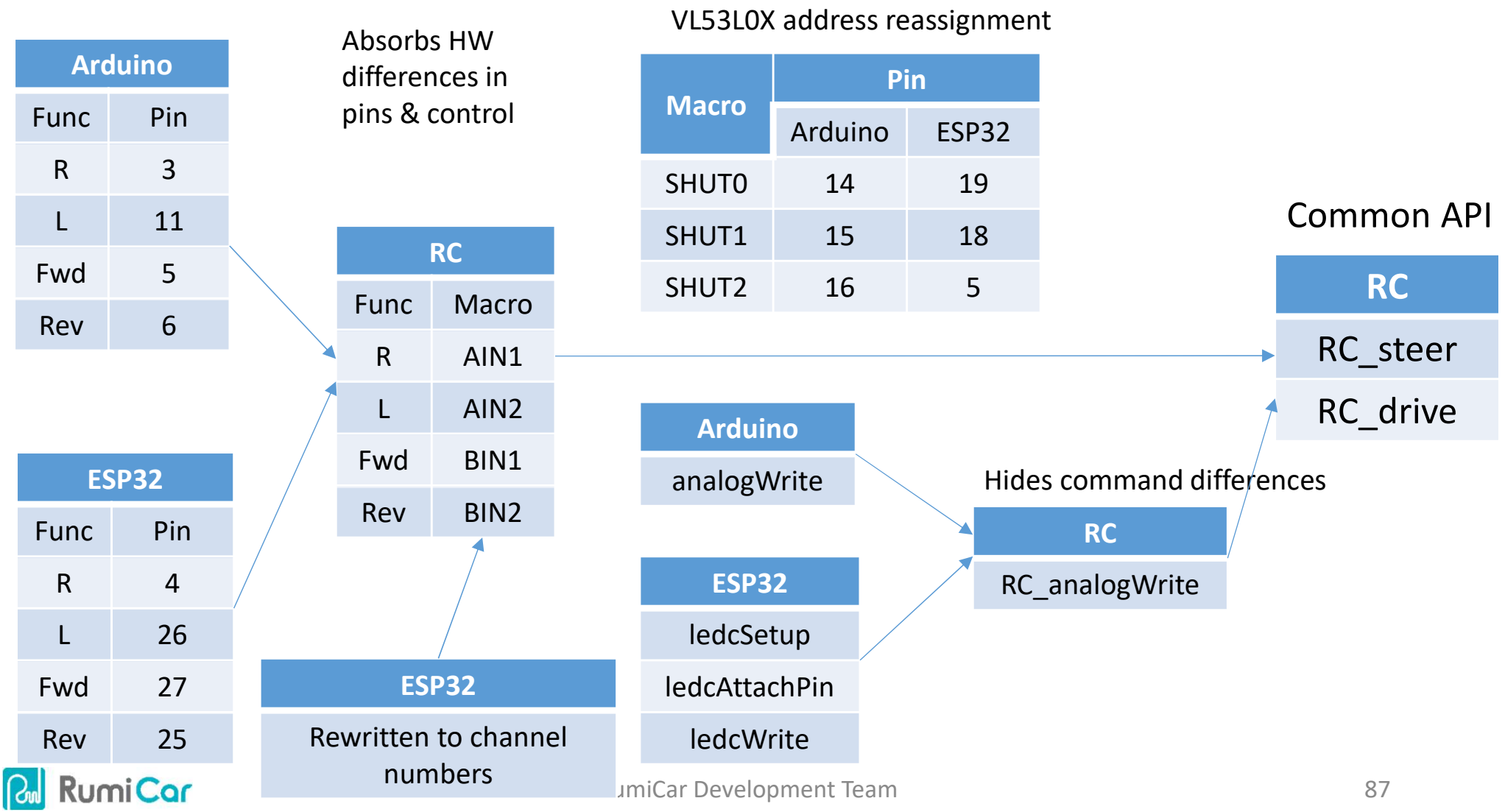
© RumiCar Development Team

Why Provide Functions

- RumiCar's macros and RC functions offer these benefits
  - Grouping commands makes programs cleaner and more readable
  - Omitting fixed arguments makes programs cleaner
  - Fewer mistakes by omitting unneeded code
  - No need to handle hardware-specific pins or control
  - Absorbs CM differences so programs can be shared

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Common Structure

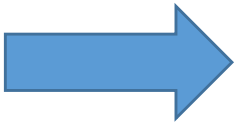


Using the RC_steer Function

- An example of steering programming using the RC_steer function
- The left side shows the code written with the original commands

Code using original commands

```
 8  void loop()
 9  {
10    analogWrite(AIN1,255);
11    analogWrite(AIN2,0);
12    delay(500);
13    analogWrite(AIN1,0);
14    analogWrite(AIN2,255);
15    delay(500);
16  }
```



Code using RC functions

```
 8  void loop()
 9  {
10    RC_steer(RIGHT);
11    delay(500);
12    RC_steer(LEFT);
13    delay(500);
14  }
```


© RumiCar Development Team

Exercise-2

Motor Control

© RumiCar Development Team

Exercise-2.1

Steering

© RumiCar Development Team

Exercise-2.1
Steering (Prep)

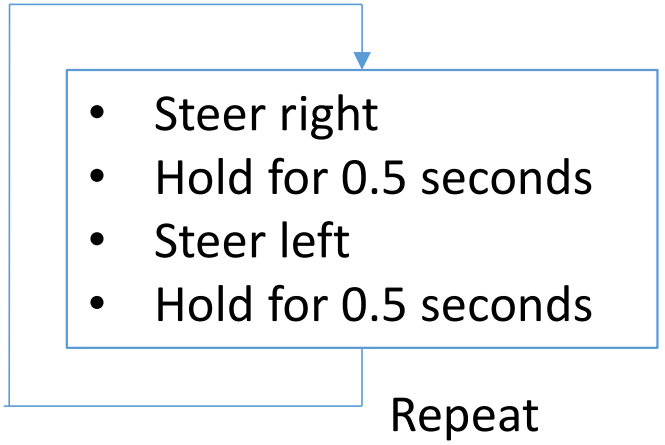
- Turn the RumiCar's power off
- Remove the CM from the RumiCar body
- Connect the USB cable to the CM
- Connect the USB cable to your computer
- Open "Exercise-2.1" from the Examples

© RumiCar Development Team

Exercise-2.1
Steering (Program)

```
 8  void loop()
 9  {
10    RC_steer(RIGHT);
11    delay(500);
12    RC_steer(LEFT);
13    delay(500);
14  }
```

- The delay command just waits for the specified time, doing nothing
- The delay argument is in milliseconds
- This program holds the steering in one direction for 0.5 seconds (500 milliseconds)



© RumiCar Development Team

Exercise-2.1
Steering (Test)

- Compile and upload the program to the CM
- Remove the USB cable from the CM
- Attach the CM to the RumiCar body
- Turn the RumiCar's power on
- Is the steering turning left and right?

© RumiCar Development Team

Exercise-2.2

Speed Control

© RumiCar Development Team

Exercise-2.2
Speed Control (Prep)

- Turn the RumiCar's power off
- Remove the CM from the RumiCar body
- Connect the USB cable to the CM
- Connect the USB cable to your computer
- Open "Exercise-2.2" from the Examples

© RumiCar Development Team

Exercise-2.2
Speed Control (Program)

```
 8  void loop()
 9  {
10    RC_drive(FORWARD,255);
11    delay(500);
12    RC_drive(FORWARD,200);
13    delay(500);
14    RC_drive(FORWARD,150);
15    delay(500);
16  }
```

- The 2nd argument of RC_drive, such as 255, 200 or 150, is the PWM value
- 255 is the maximum; 128 is about half of the maximum
- Lowering the value makes the wheels turn more and more slowly

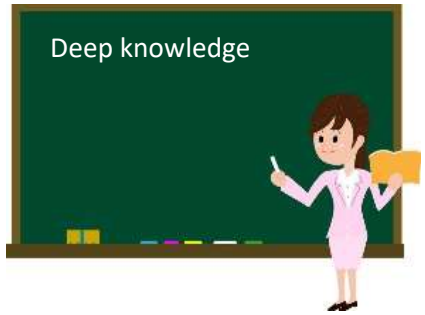
© RumiCar Development Team

Exercise-2.2
Speed Control (Test Run)

- Remove the USB cable from the CM
- Attach the CM to the RumiCar body
- First, don't let it run right away; keep holding the RumiCar body up in your hand
- Turn the RumiCar's power on
- Is the wheel speed changing?

- Gently place the RumiCar on the desk

© RumiCar Development Team



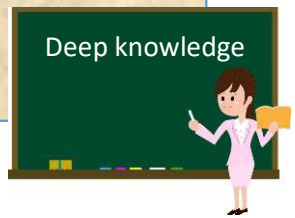
Deep knowledge

Low-Speed Control of DC Motors

Mr. Shigeki INADAMA

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RumiCar Motor Control



- The RumiCar controls two motors.
  - Steering

```
//操舵の関数
int RC_steer (int direc ){
```

```
RC_analogWrite(AIN1,255);
RC_analogWrite(AIN2,0);
```

On/off control
Full power on

- Angle can't be set
- Too much current

  - Drive

```
//走行の関数
int RC_drive(int direc, int ipwm){
```

```
RC_analogWrite(BIN1,0);
RC_analogWrite(BIN2,ipwm);
```

PWM control
Voltage is varied

Speed is unstable
Speed not linear in V

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RumiCar Motor Control



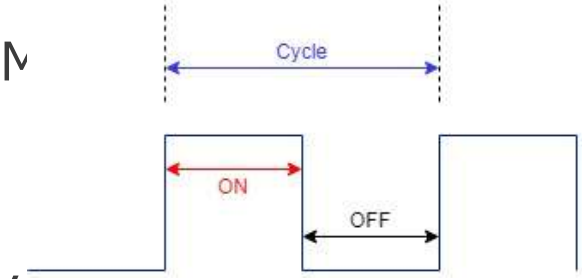
- Want low speed, but it won't move at 120/255 or below
  - Cause 1: the PWM frequency is too high
  - Cause 2: DC motors need high starting current

    $Z=j\omega L$ ... (1) $\omega=2\pi f$ ... (2)

    ≒ High apparent impedance: low PWM

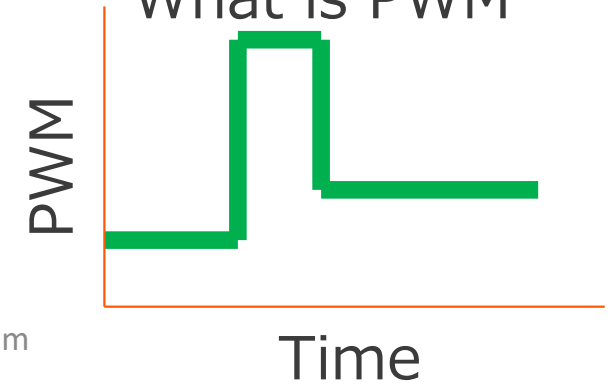
    So the usable PWM range is narrow.





Duty = ON / Cycle
VV = Duty
VF = Cycle

What is PWM



  - Fix 1: change the PWM frequency
    - 976kHz → 244Hz
  - Fix 2: introduce a start boost
    - Large PWM for the first few tens of ms

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Advanced Software Explained



- A2-3 Driving at low speed

```
//  TCCR0B |= B00000011;                    // 64分周=  16M/  64/256=976Hz  #default
  TCCR0B |= B00000100;                    // 256分周= 16M/ 256/256=244Hz
```

```
//  始動後に低速へ
  RC_drive(FORWARD,200);     RC_delay(20);
  RC_drive(FORWARD,64);      RC_delay(500);
```

Lower PWM frequency
→ current flows at low values
Strong at start, then weaker
→ now moves down to ~40

- B2-3 Low-speed operation in the library

```
// Private Variable
#define   UP_SPEED      (200)        // StartUP speed
#define   UP_TIME       (3)          // 30[ms]
```

```
RC_drive(FORWARD,40);      RC_delay(1000);
RC_drive(BRAKE,255);       RC_delay(500);
RC_drive(REVERSE,40);      RC_delay(1000);
RC_drive(BRAKE,255);       RC_delay(500);
```

PWM driver on timer interrupt
Start boost automated
→ easy low-speed driving

- B2-5 Parallel parking challenge

See it in the video!

Note: advanced programs change the PWM freq.; use RC_delay()

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Drive Motor Control in the Library

- RC_drive in RumiCar library v0.2.0 and later
  - The drive motor is controlled with slow decay (switching between drive and brake), making it easier to start even at low PWM values
  - A short, strong "start kick" is added automatically only at the moment of starting from a stop or from the opposite direction
  - Timer0 (the PWM frequency) is not changed, so delay() remains accurate
  - RC_drive(BRAKE, …) stops reliably regardless of the 2nd argument

- Advanced (Exercise-A2.3, A3.1, B2.3 to B2.5)
  - These are standalone programs that do not use the library. Use RC_delay() as explained on the previous page

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Exercise-2.3

Forward and Reverse

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Exercise-2.3
Forward & Reverse (Prep)

- Turn the RumiCar's power off
- Remove the CM from the RumiCar body
- Connect the USB cable to the CM
- Connect the USB cable to your computer
- Open "Exercise-2.3" from the Examples

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Exercise-2.3
Forward & Reverse (Program)

- Moves forward when the 1st argument of RC_drive is FORWARD
- Moves backward when the 1st argument is REVERSE

```
 8  void loop()
 9  {
10    RC_drive(FORWARD,255);
11    delay(500);
12    RC_drive(REVERSE,255);
13    delay(500);
14  }
```

- Go forward
- Hold for 0.5 seconds (keep going forward)
- Go backward
- Hold for 0.5 seconds (keep going backward)

Repeat

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Exercise-2.3
Forward & Reverse (Test Run)

- Compile and upload the program to the CM
- Remove the USB cable from the CM
- Attach the CM to the RumiCar body
- First, don't let it run right away; keep holding the RumiCar body up in your hand
- Turn the RumiCar's power on
- Are the wheels repeatedly going forward and backward?

- Gently place the RumiCar on the desk

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Exercise-2.3
Forward & Reverse (Challenge)

- (Advanced question) How can you repeat forward and reverse while slipping as little as possible?

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Exercise-2.4

Zig-Zag Driving

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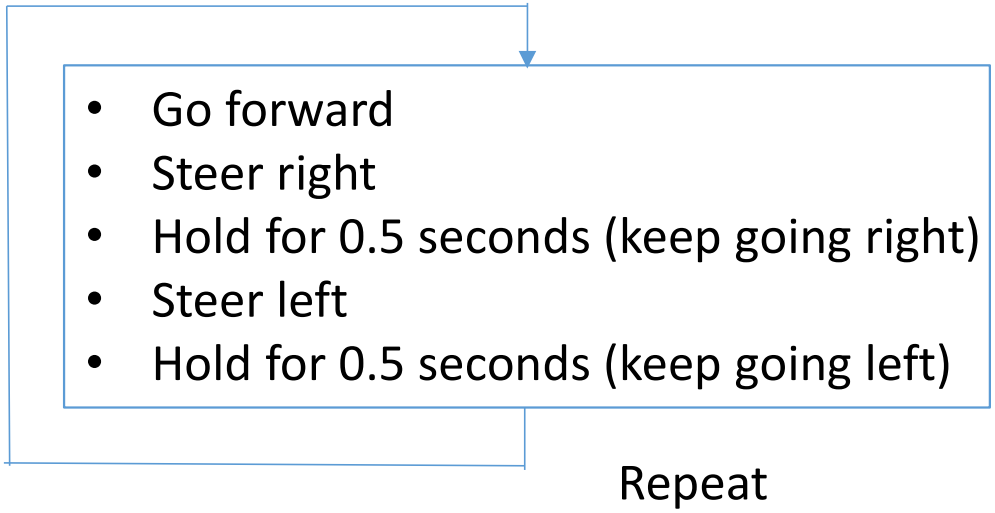
Exercise-2.4
Zig-Zag Driving (Prep)

- Turn the RumiCar's power off
- Remove the CM from the RumiCar body
- Connect the USB cable to the CM
- Connect the USB cable to your computer
- Open "Exercise-2.4" from the Examples

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Exercise-2.4
Zig-Zag Driving (Program)

```
 8  void loop()
 9  {
10    RC_drive(FORWARD,255);
11    RC_steer(RIGHT);
12    delay(500);
13    RC_steer(LEFT);
14    delay(500);
15  }
```



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Exercise-2.4
Zig-Zag Driving (Test Run)

- Compile and upload the program to the CM
- Remove the USB cable from the CM
- Attach the CM to the RumiCar body
- First, don't let it run right away; keep holding the RumiCar body up in your hand
- Turn the RumiCar's power on
- Is the RumiCar moving as you expected?

- Gently place the RumiCar on the desk

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Autonomous Driving Basics

Controlling the Car with Sensors

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Autonomous Driving Basics

- Autonomous driving means making a car drive automatically and optimally using information from sensors, information it already has, or a combination of them
- In this chapter, we will consider a simple example as the basics of autonomous driving

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Basics: Assumed Situation

- Assume the following simple autonomous driving situation
  - While driving forward, if an obstacle is detected ahead, stop safely without colliding with it
  - While stopped after detecting an obstacle, resume driving forward when the obstacle is gone

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Exercise-3

Let's Develop an Autonomous Driving Program!

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Exercise-3.1

Stopping the Car Safely

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Exercise-3.1
Stopping the Car Safely

- Specifically, assume the following simple conditions
  - While detecting obstacles with the RumiCar's central sensor (Sensor1), drive forward when no obstacle is detected
  - Stop, for example, 10cm before the obstacle, and resume driving forward when the obstacle is gone

Exercise-3.1
RumiCar Development Notes

- Distance measurement error
  - In high-speed mode, the VL53L0X setting in the sample programs, some accuracy is sacrificed to measure quickly. Therefore an error of 1-2cm occurs when ranging. Allow a margin when judging distance

- Handling failed measurements
  - A measurement of around 8190 means no laser reflection was obtained, i.e. no obstacle could be detected within the measurement range
  - (RC_read returns -3 in this case)

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Exercise-3.1
RumiCar Development Notes (Cont'd)

- Detecting the ground, etc.
  - The sensor's laser is emitted in a cone. Because the RumiCar is small, the sensors are mounted low, so the cone-shaped laser eventually hits the ground or floor before the measurement limit. This distance is about 30cm on the RumiCar model 32. For these reasons, set the obstacle detection distance to 30cm or less

- Sudden stops
  - Even if you command a stop or reverse while driving, the car cannot stop immediately because of inertia (momentum). If possible, think about good ways to slow down and stop

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Exercise-3.1
Programming Requirements

Item	Condition
Sensor used	Use only the central sensor (Sensor1)
Behavior	Stop when an obstacle is detected within the set distance
Set distance	10cm. Considering error, the actual range is 10 ± 2cm
Reliable distance	Within 30cm
Value when measurement fails	8190, but note it may contain errors (-3 with RC_read)

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Exercise-3.1
A Development Tip

- If the condition is simply "no obstacle", the RumiCar will drive on forever when there is no obstacle ahead. Then, if you carelessly put the RumiCar on a desk, it will jump off the desk. If you may test it on a desk, for safety it is a good idea to add a condition such as driving only when there is an obstacle within 30cm. This reduces the risk of accidentally damaging the RumiCar during development

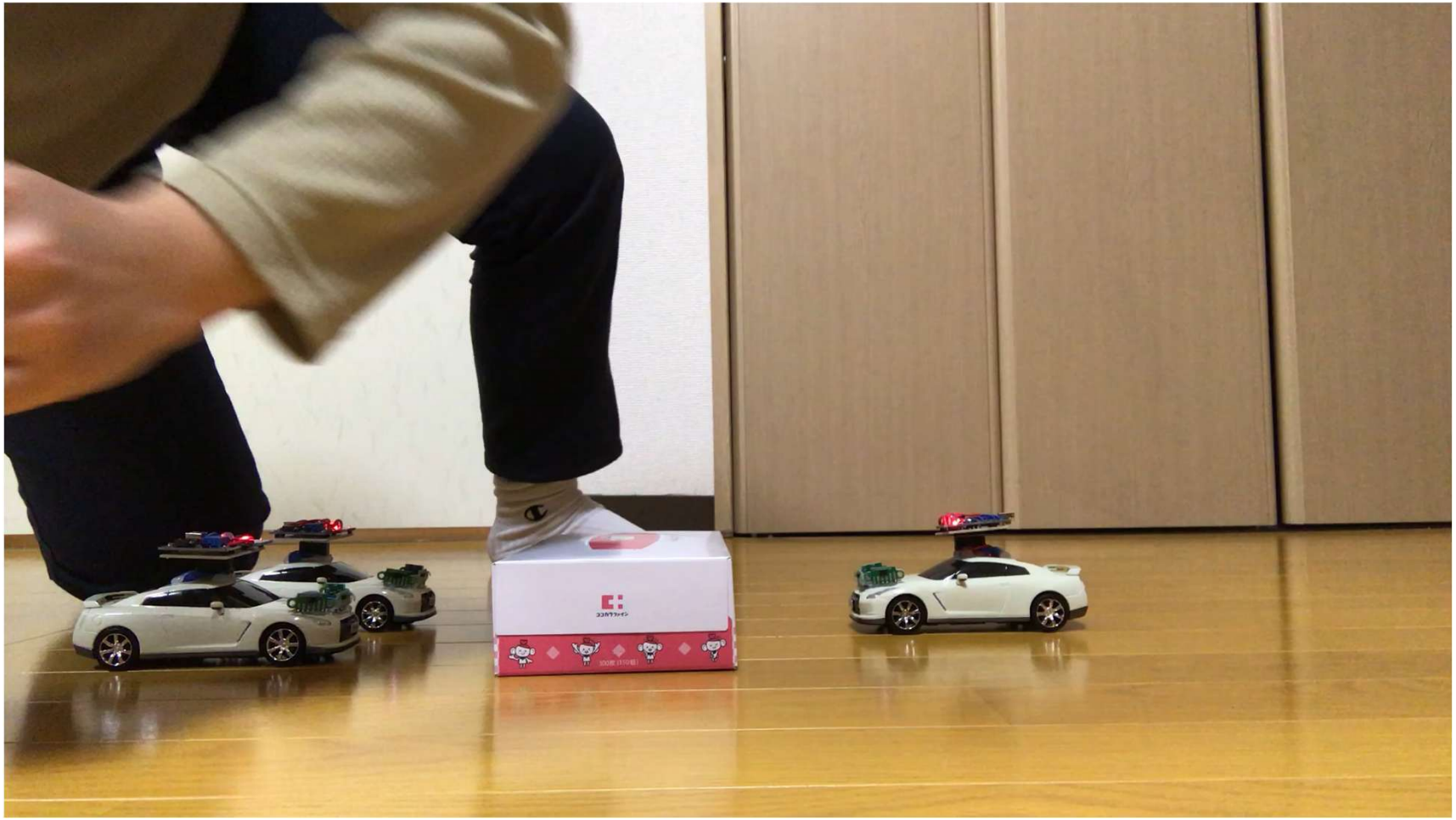
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Exercise-3.1 https://youtu.be/95pc_4Wf14U
Safe Stop (Video)



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Exercise-3.1 https://youtu.be/95pc_4Wf14U
Safe Stop (Video)



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Exercise-3.1
Safe Stop (Prep)

- Turn the RumiCar's power off
- Remove the CM from the RumiCar body
- Connect the USB cable to the CM
- Connect the USB cable to your computer
- Open "Exercise-3.1" from the Examples

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Exercise-3.1
Safe Stop (Program)

```
 8  void loop()
 9  {
10
11    int ispeed = 255;
12    int idist1;
13    idist1=sensor1.readRangeSingleMillimeters();
14    Serial.println(idist1);
15    if ( idist1 < 300 ){
16      if ( idist1 > 120 ){
17        RC_drive(FORWARD,ispeed);
18      }else if (idist1 < 80){
19        RC_drive(REVERSE,ispeed);;
20      }else{
21        RC_drive(BRAKE,ispeed);
22      }
23    }else{
24      RC_drive(FREE,ispeed);
25    }
26  }
```

- Is there an obstacle within 30cm?
- (If not, do nothing)
- (If yes, do the following)
  - Obstacle farther than 12cm -> forward
  - Obstacle closer than 8cm -> reverse
  - 8cm to 12cm -> brake

Repeat

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Exercise-3.1
Safe Stop (Test Run)

- Compile and upload the program to the CM
- Remove the USB cable from the CM
- Attach the CM to the RumiCar body
- Turn the RumiCar's power on
- Move your hand toward and away from the central sensor. Is the RumiCar moving as you expected?

- Gently place the RumiCar on the desk

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Exercise-3.2

Driving in Urban Areas

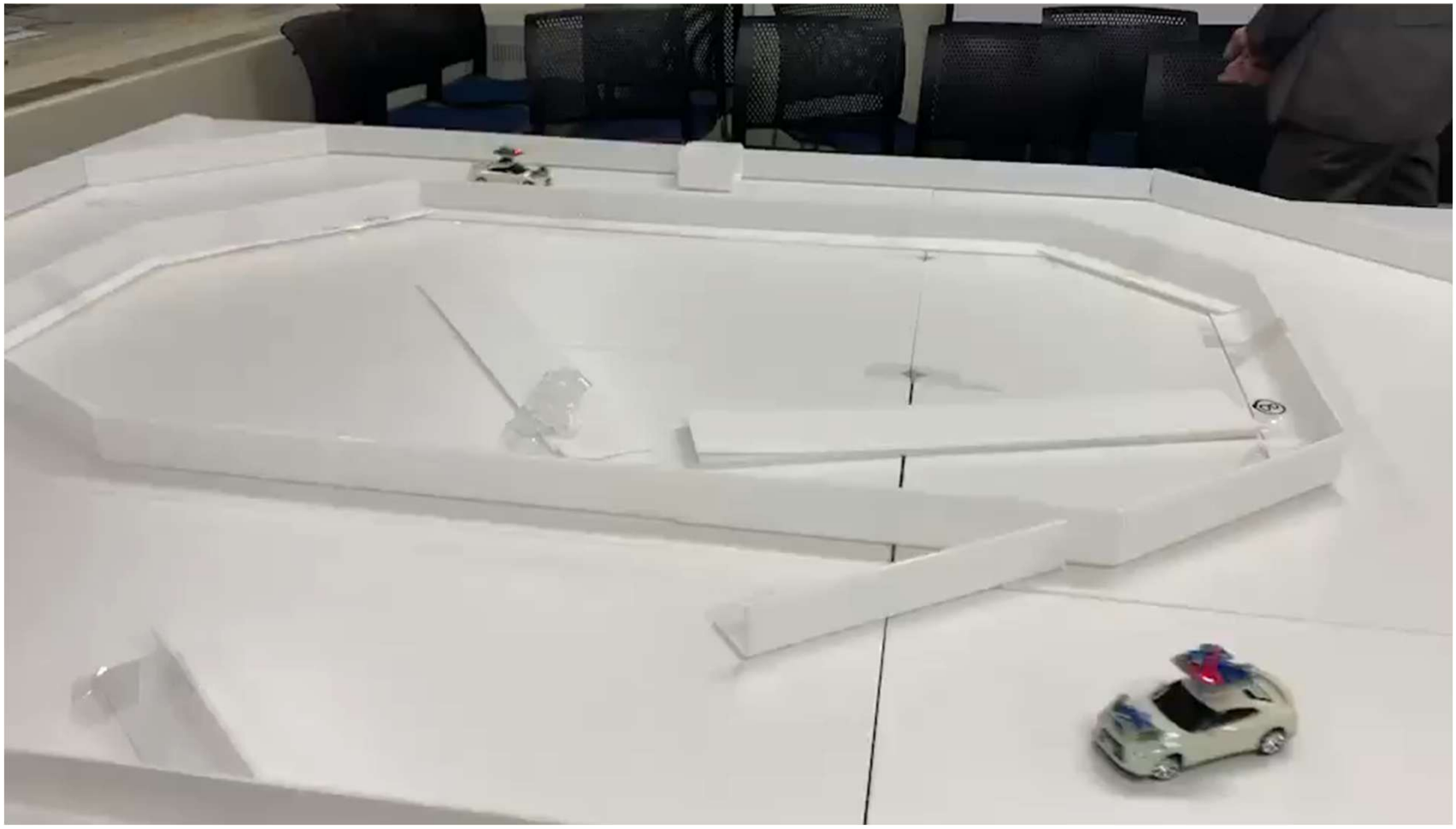
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Exercise-3.2
Driving in Urban Areas

- Specifically, assume the following simple conditions
  - Drive on a road with walls on both sides, representing an urban area
  - Measure the distances to the walls on both sides and drive in the center of the road
  - Stop, for example, 10cm behind the car ahead, and resume driving forward when the car ahead is gone

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Exercise-3.2 https://youtu.be/jyq4Ph0mCDM
Urban Driving (Video)



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Exercise-3.2
Urban Driving (Prep)

- Turn the RumiCar's power off
- Remove the CM from the RumiCar body
- Connect the USB cable to the CM
- Connect the USB cable to your computer
- Open "Exercise-3.2" from the Examples

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Exercise-3.2
Urban Driving (Program)

```
 8  void loop()
 9  {
10
11    int ibound =250;
12    int s0, s1, s2;
13    s0=sensor0.readRangeSingleMillimeters();
14    s1=sensor1.readRangeSingleMillimeters();
15    s2=sensor2.readRangeSingleMillimeters();
16
17
18    Serial.print("Sensor0:");
19    Serial.print(s0);
20    Serial.print("  Sensor1:");
21    Serial.print(s1);
22    Serial.print("  Sensor2:");
23    Serial.println(s2);
24
25    if(s1<100){
26      RC_drive(REVERSE,150);
27    }else if (s1<150){
28      RC_drive(FORWARD,150);
29    }else if (s1<250){
30      RC_drive(FORWARD,200);
31    }else{
32      RC_drive(FORWARD,255);
33    }
34    if(s0>s2){
35      RC_steer(LEFT);
36    }else{
37      RC_steer(RIGHT);
38    }
39  }
```

- Try other speed values (150, 200, 255)!
- Try other distance values (100, 150, 250)!
- What will happen??

- Obstacle closer than 10cm -> slowly reverse
- Obstacle closer than 15cm -> slowly forward
- Obstacle closer than 25cm -> a bit faster forward
- Otherwise -> full speed forward

- Compare the distances to the left and right walls
- Move away from the closer wall (steer toward the farther wall)

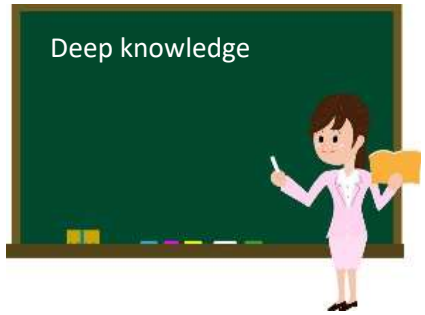
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Exercise-3.2
Urban Driving (Test Run)

- Compile and upload the program to the CM
- Remove the USB cable from the CM
- Attach the CM to the RumiCar body
- Turn the RumiCar's power on

- Gently place the RumiCar on the course

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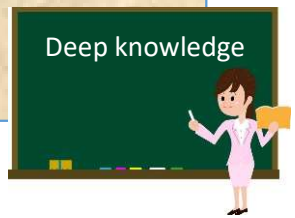
Deep knowledge

Raspberry Pi Sample Programs

Ms. Shoko ISHIGAKI

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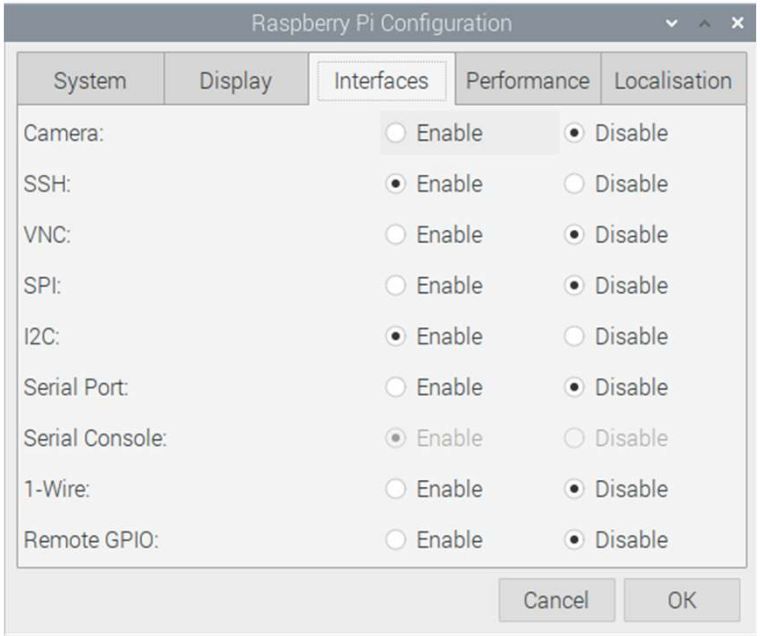
Raspberry Pi Sample Programs



- The pigpio library is used for all GPIO control, including PWM control. Start pigpiod before running the program.

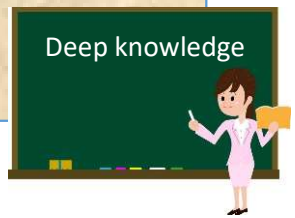
  Start command: sudo pigpiod

- The ranging modules use I2C, so enable I2C beforehand.



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Raspberry Pi Sample Programs



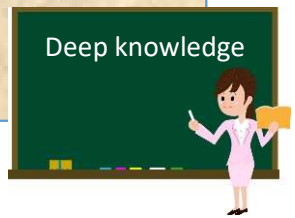
- Software PWM and hardware PWM
  - The Raspberry Pi has two independently controllable hardware PWMs

  Pins 12/32 and pins 33/35 can be used as hardware PWM

  - Software PWM can be used on all GPIO pins
  - Hardware PWM is used for the drive motor, which needs precise control such as sudden stops

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Raspberry Pi Sample Programs



How to install the library needed to run the VL53L0X program

- Use the library published on GitHub

1. git clone https://github.com/johnbryanmoore/VL53L0X_rasp_python
2. Run make in the cloned VL53L0X_rasp_python folder

Notes

- The samples assume the repository is cloned to /home/pi/Documents. Change the shared library path in VL53L0X.py if needed.
- If the VL53L0X Vcc is not the RasPi's 3.3V output, connect the RasPi GND to the VL53L0X GND

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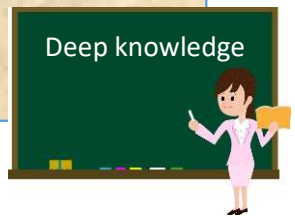
Azure Iot Hub



- Two-way communication between IoT devices (Raspberry Pi, etc.) and Azure
- Enables visualization of driving data and remote control
- Integration with other Azure services such as Azure Machine Learning

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Device-to-cloud messages

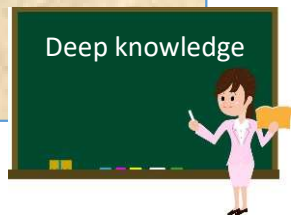


- Sending and monitoring sensor data
- Can send data up to 256KB

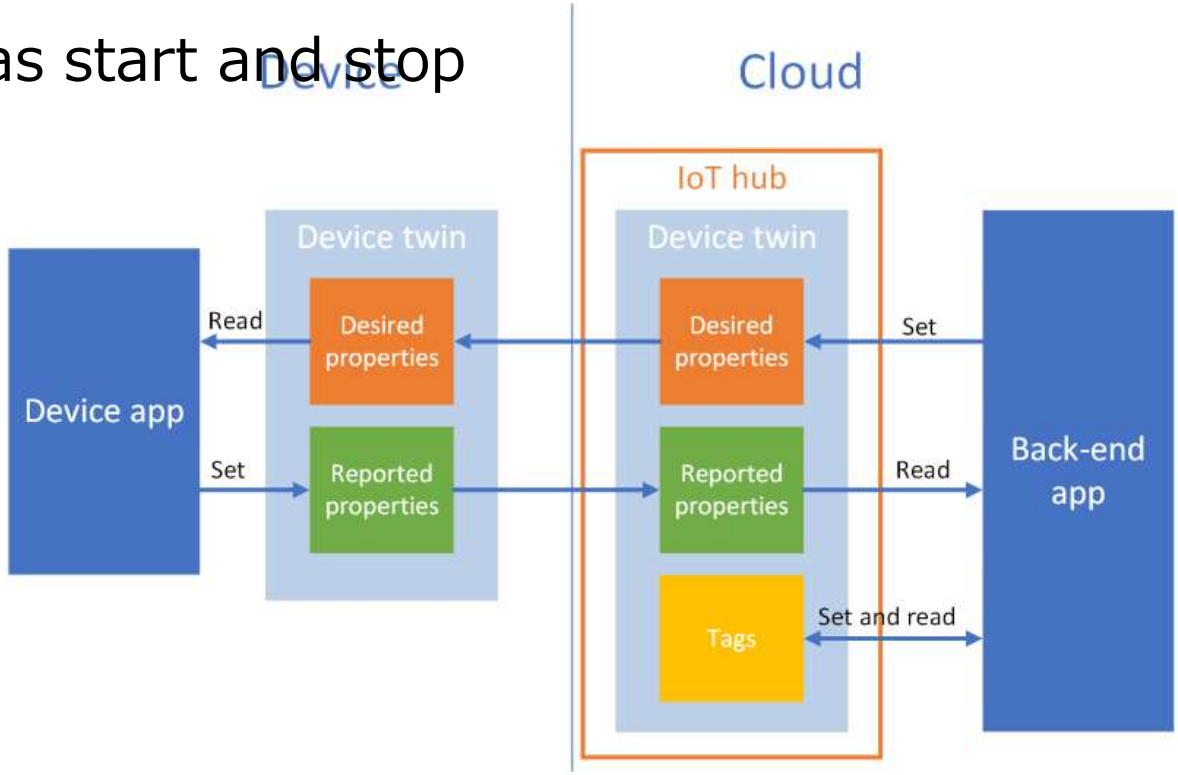
```
ShokonoMacBook-Pro:~ $ az iot hub monitor-events --hub-name raspi-ishigaki --device-id raspi
Starting event monitor, filtering on device: raspi, use ctrl-c to stop...
{
    "event": {
        "origin": "raspi",
        "interface": "",
        "payload": "{\"distance\": 424, \"control\": FREE}"
    }
}
{
    "event": {
        "origin": "raspi",
        "interface": "",
        "payload": "{\"distance\": 422, \"control\": FREE}"
    }
}
{
    "event": {
        "origin": "raspi",
        "interface": "",
        "payload": "{\"distance\": 422, \"control\": FREE}"
    }
}
```


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desired properties



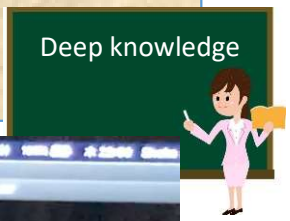
- Cloud-to-device notifications
- Control commands such as start and stop

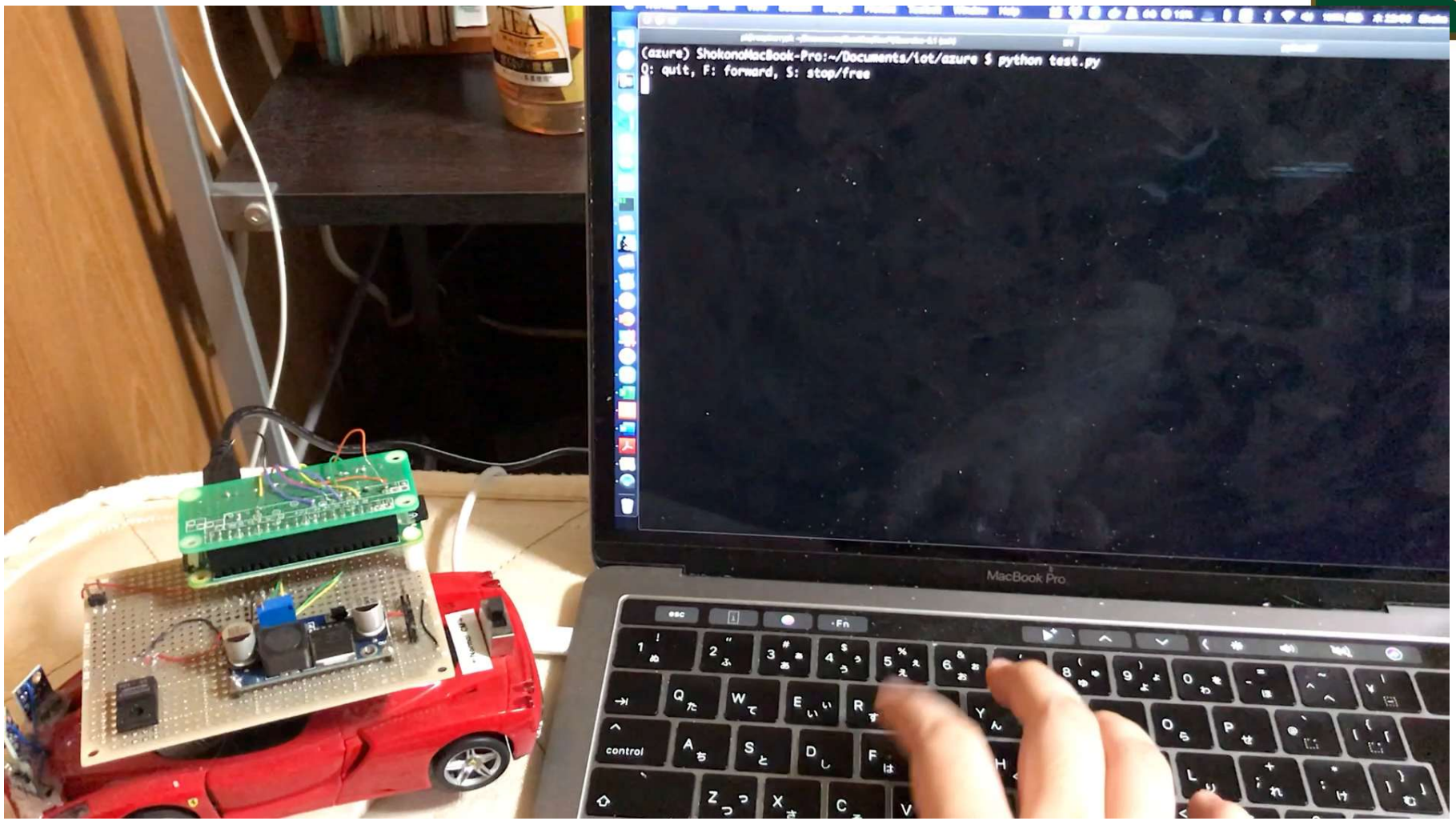


https://docs.microsoft.com/en-us/azure/iot-hub/tutorial-device-twins

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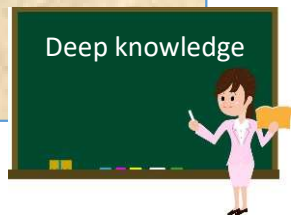
Operation Video (Video)





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References



- Azure IoT Hub tutorials
  - https://docs.microsoft.com/ja-jp/azure/iot-hub/quickstart-send-telemetry-cli
- Source code (Raspberry Pi)
  - https://github.com/RumiCar-group/RumiCar/tree/master/RasPi/misc
  - Sending to the cloud: azure_d2c.py
  - Desired property: azure_property.py (Raspberry Pi side), azure_service.py (controlling side, e.g. PC)

*You need to create an Azure IoT Hub.

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Appendix

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I2C Addresses

- RumiCar uses I2C communication. These are the addresses currently in use

Sensor			Address	How to set
VL53L0X (Laser ranging module)	Sensor0	Left	0x14	Changed by software
	Sensor1	Center	0x15	
	Sensor2	Right	0x16	
	Sensor3	Rear	0x17	Mounted on the rear of the CM (optional). Changed from 0x29 at startup
BMX055 (9-axis sensor)	Accelerometer		0x19	Jumper setting on the board
	Gyro sensor		0x69	
	Magnetic sensor		0x13	

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Contributors by Area

Name	Details
Hideaki KATAOKA	• RumiCar car body building • CM development
Yutaka OHE	• Sensor PCB design and development • Arduino Nano CM PCB design and development
Shoko ISHIGAKI	• Raspberry Pi CM development • Python program development • Microsoft Azure IoT Hub integration
Shigeki INADAMA	• Motor low-speed control • ESP32 PCB design and development • M5StickC PCB design and development • Sensor PCB design and development
Hideki TAKASE	• Library development
Yoshiteru CHIBA	• Arduino official library registration and maintenance

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Event History

	Date	Event	Venue	Notes
1	August 5-6, 2017	MakerFaireTokyo2017	Tokyo Big Sight	
2	May 18-20, 2018	MakerFaireBayArea2018	San Mateo, USA	
3	August 4-5, 2018	MakerfaireTokyo2018	Tokyo Big Sight	
4	March 15, 2019	Hello Tomorrow Global Challenge	Paris, France	Investor event
5	August 3-4, 2019	MakerFiareTokyo2019	Tokyo Big Sight	
6	November 30, 2019	IoT ALGYAN Tsukuruyo 2!	Joho Kagaku Senmon Gakko (Information Science College)	
7	April 18, 2020	IoT ALGYAN RumiCar Hands-On (live stream)	Online	
8	April 25, 2020	IoT ALGYAN RumiCar Hands-On (live stream)	Online	
9	June 6, 2020	IoT ALGYAN RumiCar Hands-On	Online	Online with lent cars

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Event History

	Date	Event	Venue	Notes
10	June 20, 2020	[In person] My RumiCar Test Run! Run your favorite machine together with RumiCar Development Team members!	Asao Ward, Kawasaki	
11	August 10, 2020	RumiCar Hands-On with Real Cars & Running Meet	Yokohama Port Opening Memorial Hall	
12	August 30, 2020	RumiCar Running Meet	Asao Culture Center	
13	November 13, 2020	- Learn the Basics of Self-Driving Car Control with RumiCar - Session 1: Parts check, building the hardware	Zoom	Organizer: Technical Committee, Hiroshima Information Industry Association Sponsor: RumiCar Development
14	December 4, 2020	- Learn the Basics of Self-Driving Car Control with RumiCar - Session 2: Continued hardware building, basic software tests		

Event History

	Date	Event	Venue	Notes
15	January 10, 2021	15th Event: RumiCar Hands-On & Running Meet (held in a 300-person venue!)	Asao Ward, Kawasaki	
16	September 26, 2021	RumiCar Conference	Online	
17	November 28, 2021	RumiCar Running Meet ("Good Kitty Day")	Asao Culture Center	
18	December 12, 2021	RumiCar Conference 12/12	Online	YouTube Live
19	April 29, 2022	Makers' Flea Market	Giken Base	
20	August 7, 2022	17th Event: RumiCar Test Run	Shinagawa City, Tokyo Osaki No.2 Community Hall	Event chair: Mr. Inoue
21	September 3-4, 2022	Maker Faire Tokyo2022	Tokyo Big Sight	

Event History

	Date	Event	Venue	Notes
23	November 3, 5, 6, 12 and 13, 2022	2nd RumiCar Gallery (solo exhibition)	Sumida Park Riverside Gallery	
24	December 31, 2022	Comic Market 101	Tokyo Big Sight	
25	February 1-3, 2023	Technical Show Yokohama 2023	PACIFICO Yokohama	Exhibited at the R2 booth
26	March 25, 2023	Self-Driving Algorithm Platform "RumiCar" Lecture & Hands-On	Sony City Osaki BRIDGE TERMINAL	
27	July 19-21, 2023	Robot Innovation Week 2023	Nagoya Congress Center	Exhibited at the R2 booth
28	October 14-15, 2023	Maker Faire Tokyo 2023	Tokyo Big Sight	

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Event History

	Date	Event	Venue	Notes
29	October 26-27, 2023	Tsubame-Sanjo Monozukuri Messe 2023	Tsubame-Sanjo Regional Industries Promotion Center	RumiCar Niigata exhibited the Taiguruma (sea-bream cart) version
30	November 3-4, 2023	HANAZONO EXPO 2023	Hanazono Central Park (Higashiosaka)	Exhibited at the R2 booth
31	November 26, 2023	Self-Driving Mini Car Battle 2023, Elementary School Division	Sky Hall Toyota	RumiCar used for the car body (Toyota Engineering Society)
32	November 29 - December 2, 2023	International Robot Exhibition 2023	Tokyo Big Sight	Exhibited in the R2 area

Event History

	Date	Event	Venue	Notes
34	December 16, 2023	BLACK TECH SATURDAYS	Detroit, USA	Exhibited by Detroit Autonomous Vehicle Group
35	January 21, 2024	2nd Tobitate Festival	MEXT East Building	Exhibition and on-stage talk
36	February 3-4, 2024	YOXO FESTIVAL 2024	Minato Mirai, Yokohama	Exhibition and workshop; reached the pitch finals

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Event History

	Date	Event	Venue	Notes
37	February 24, 2024	[Toyota Engineering Society presents] Let's Build a Self-Driving Mini Car!	SMBC Park Sakae (Nagoya)	Workshop held by Toyota Engineering Society
38	March 2-3, 2024	Kariya Micro Maker Faire 2024	Kariya City Industrial Promotion Center	Exhibited at the SPRESENSE booth
39	May 20, 2024	Driving the Future (AutoSens USA 2024)	Detroit, USA	Used in a hands-on for elementary school students
40	July 6-7, 2024	Tech Seeker 2024 Board Computer Programming Class	ATC (Osaka)	Workshop with the SPRESENSE version

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Event History

	Date	Event	Venue	Notes
41	July 2024	Let's Move a Car with "Self-Driving" Using Sensors! (class)	Shibaura Institute of Technology Junior and Senior High School	Class with the SPRESENSE version
42	July 24, 2024	Same as above: final race	Sony Headquarters	Time attack
43	September 21-22, 2024	Maker Faire Tokyo 2024	Tokyo Big Sight	
44	September 23, 2024	RoboRAVE Kobe 2024	Kobe International House	
45	October 6, 2024	Yuru Mechatro Exhibition 2	Tokyo Technical College	RumiCar Niigata exhibited the Taiguruma (sea-bream cart) version
46	October 18-20,	Maker Faire Bay Area 2024	Vallejo, USA	Received a Blue

Event History

	Date	Event	Venue	Notes
47	October 26-27, 2024	Tsukuroka! 3	Osaka Municipal Central Library	
48	October 27, 2024	Self-Driving Mini Car Battle 2024, Elementary School Division	Sky Hall Toyota	RumiCar used for the car body (Toyota Engineering Society)
49	December 14-15, 2024	Aichi Robo Fes 2024	Aichi Sky Expo	Workshop held by Toyota Engineering Society
50	FY2024	Makihigashi Junior High School "Community Circle Activities"	Niigata City Makihigashi Junior High School	
51	February 8, 2025	[Toyota Engineering Society presents] Let's Build a Self-Driving	SMBC Park Sakae	Workshop held by Toyota

Event History

	Date	Event	Venue	Notes
52	March 8, 2025	Raspberry Pi 13th Birthday Party (Raspberry JAM Tokyo 2025)	Sony City Osaki BRIDGE TERMINAL	Lightning talk
53	March 16, 2025	8th WE Love Toyota Festa	Sky Hall Toyota	Exhibited at the Self-Driving Mini Car Club booth
54	July 12, 2025	Programming class for elementary and junior high school students	FUN+TECH LABO (Sagamihara)	Workshop with the SPRESENSE version
55	July 27, 2025	RumiCar Program an Autonomous Vehicle	Cambridge, UK	Workshop held
56	November 8, 2025	Tokai Environment Festa 2025	Tokai Village Office	
57	March 28, 2026	NT Fukushima 2026	Aizu Keikodo (Aizuwakamatsu)	

Event History

	Date	Event	Venue	Notes
58	April 22, 2026	STEAM Fair 2026	Northville, USA	
59	May 10, 2026	Tech Festival	Cambridge, UK	
60	August 11, 2026	RumiCar Programming Class (morning and afternoon), 18th RumiCar Running Meet	Hokutopia (Kita City, Tokyo)	
61	August 30, 2026	RumiCar Car Body Building Class	bono Sagami-Ono South Mall	
62	September 23, 2026	RumiCar Programming Class (morning and afternoon, with an English class), 19th RumiCar Running Meet	Hokutopia (Kita City, Tokyo)	

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Media Coverage

	Media	Date	Details	Page
1	I/O Vol.47, July issue (No.549)	July 2022 issue	100-Yen Robot Club!	118-119
2	I/O Vol.47, August issue (No.550)	August 2022 issue	RumiCar	10-13
3	The World of Electronics Crafts Expanding from "Raspberry Pi"	Released November 30, 2022		

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Partnerships

	Media	Date	Details
1	R2		
2			
3			

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Final Notes

RumiCar Info

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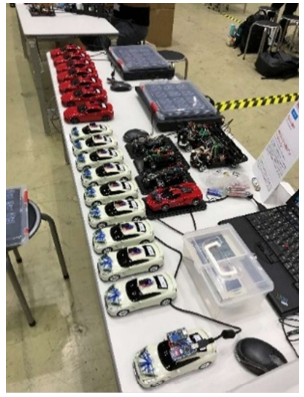
For More Enjoyment

- The RumiCar Development Team believes standardization is important so that many people can enjoy RumiCar
- Standardization improves the hardware compatibility of car bodies and compute modules and increases the reusability of programs
- This lets many people improve, add and enhance functions without the trouble of making changes
- So that more people can enjoy it, please follow the RumiCar standard for compute module pin assignments and pin layout when developing

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RumiCar Info

- RumiCar Development Team Web Page
  - https://www.rumicar.com/
- Facebook RumiCar Group
  - https://www.facebook.com/groups/rumicar
- RumiCar Development Team Contact E-mail
  - info@RumiCar.com
- RumiCar YouTube
  - https://www.youtube.com/channel/UCVg3CBSVBcc_00FdC6q2wDg
- GitHub
  - https://github.com/RumiCar-group/RumiCar
  - https://github.com/RumiCar-group/RumiCar-lib



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Thank you for participating in the RumiCar Hands-On!

We're done!

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